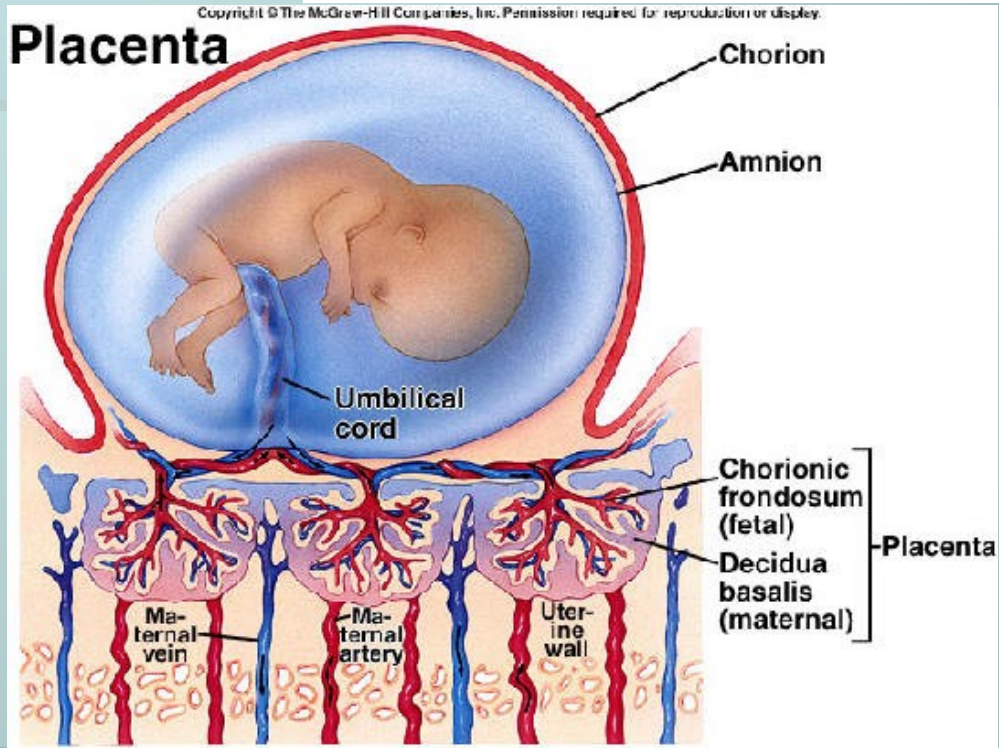
STRUCTURE OF PLACENTA

DR.SAHAR MUBEEN
ASSOCIATE PROFESSOR
DIMC







LEARNING OBJECTIVES

After completion of this lecture, students will be able to:

- DEFINE PLAENTA
- DISCUSS FORMATION OF PLACENTA
- DESCRIBE STRUCTURE OF PLACENTA
- KNOW ABOUT THE STRUCTURE OF VILLUS
- DISCUSS THE TYPES OF VILLI

PLACENTAL REVIEW

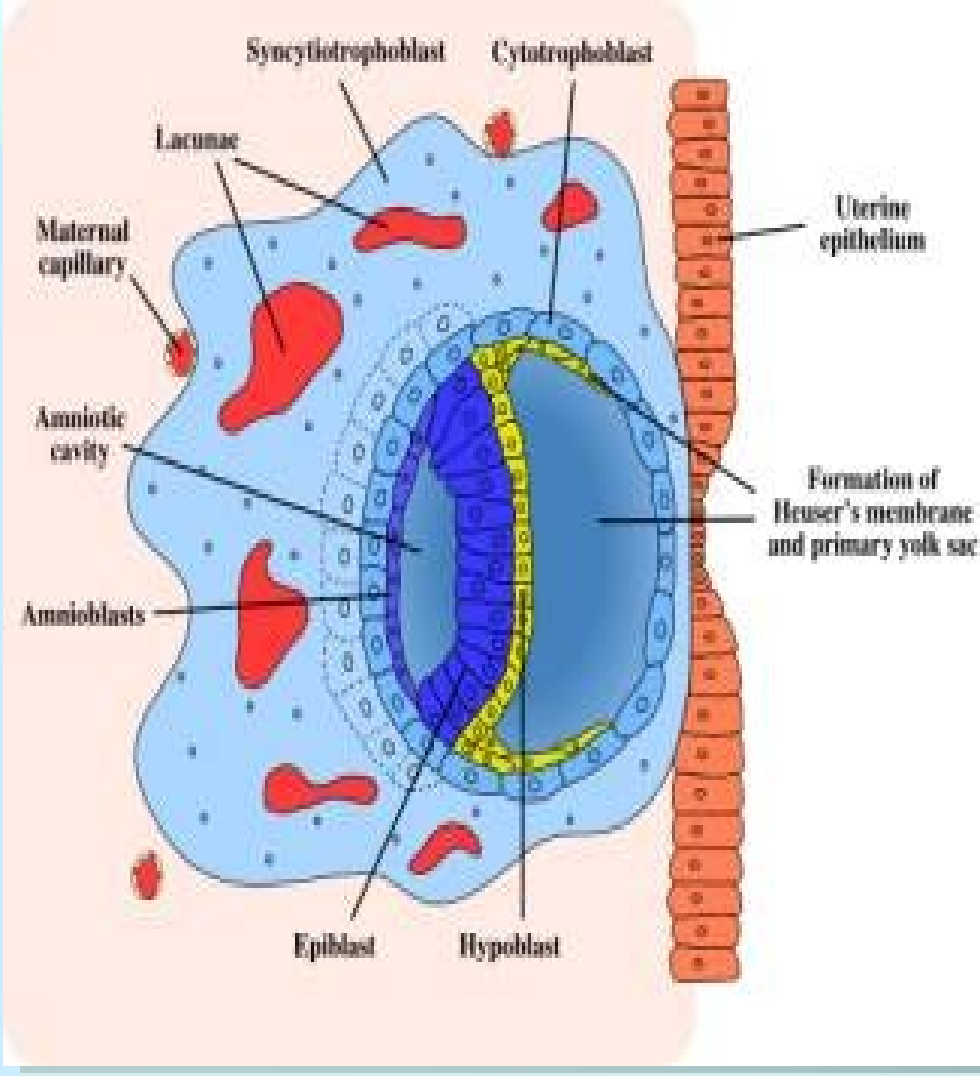
- What is placenta?
- What are its main components?
- What is decidua?
- What is chorionic villi?
- What is placental barrier?

Fill in the Blanks

1. Fetal membranes include:----------, ----------, ----------, ----------, and ----------
2. ---------- has maximum tensile strength
3. If amniotic fluid is > 2000ml, the condition is known as ----------
4. Extraembryonic coelom is also known as ----------
5. Chorion is made up of ---------- and ----------
6. Secondary yolk sac is lined by ----------
7. In about 2% of adults the proximal intra-abdominal part of yolk stalk persists as ----------
8. Remnant of allantois is known as ----------
9. Trophoblast consists of 2 layers:---------- and ----------
10. Umbilical Cord consists of mucous CT ----------, 3 vessels, yolk sac, allantois

PLACENTA

- Vital organ/life support system for fetus
- Fleshy and multi purpose organ
- Develops from **fetal chorionic tissue** and **maternal decidua** during pregnancy
- Development period: 6th - 12th wk
- Placenta and umbilical cord are a **transport system** for substances between mother and fetus



Trophoblast differentiates into syncytiotrophoblast and cytotrophoblast

PLACENTAL FUNCTION

- Function as an **efficient gas exchange membrane**, allowing oxygen and carbon dioxide to pass between two circulations, without allowing two circulations to mix
- Permeable to **nutritional substances** such as **vitamins, glucose, free fatty acids and electrolytes**
- **Passive immunity** is acquired by fetus through passage of antibodies such as those related to **diphtheria**, **smallpox** and **measles**
- **ENDOCRINE GLAND:** Produces several **hormones** including **human chorionic gonadotrophin and the steroid hormones progestins and estrogens.**

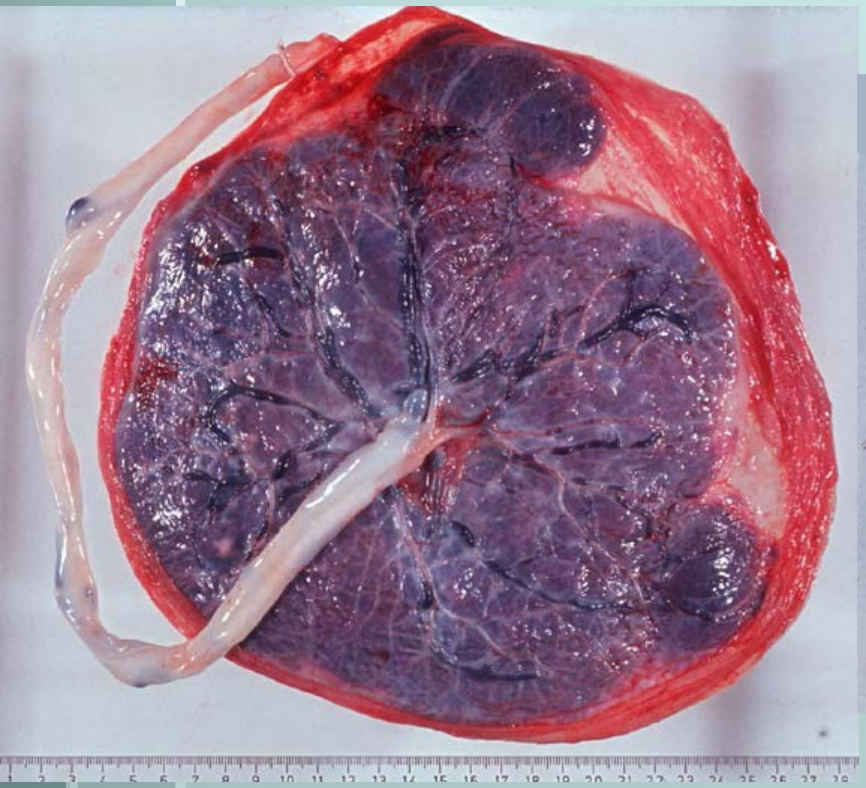
FULL TERM PLACENTA NAKED EYE ANATOMY

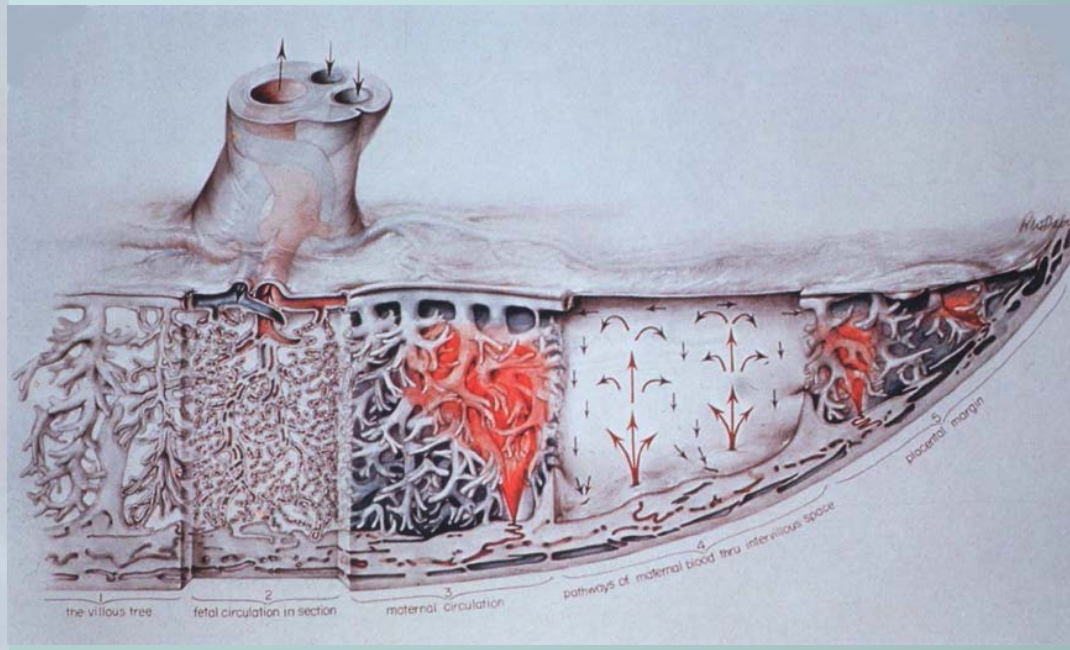
- Flattened circular shape (DISCOID)
- **Diameter**: 8 inches(20cm)
- **Thickness:** inch(2.5 cm)
- **Weight: 500- 600** grams
- Occupies 30% of uterine wall
- **Hemochorial:** A placenta in which the maternal blood is in direct contact with the chorion





FULL TERM PLACENTA
NAKED EYE ANATOMY

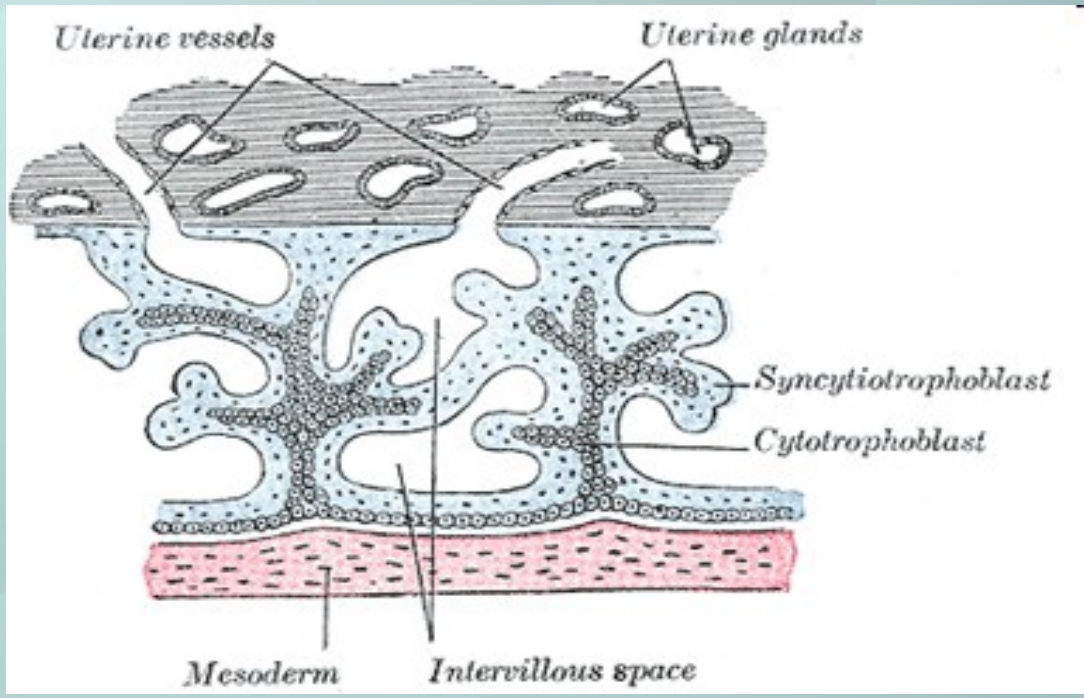




Internally it consists of a fetal villous tree bathed directly by maternal blood, at least during the 2nd and 3rd trimesters

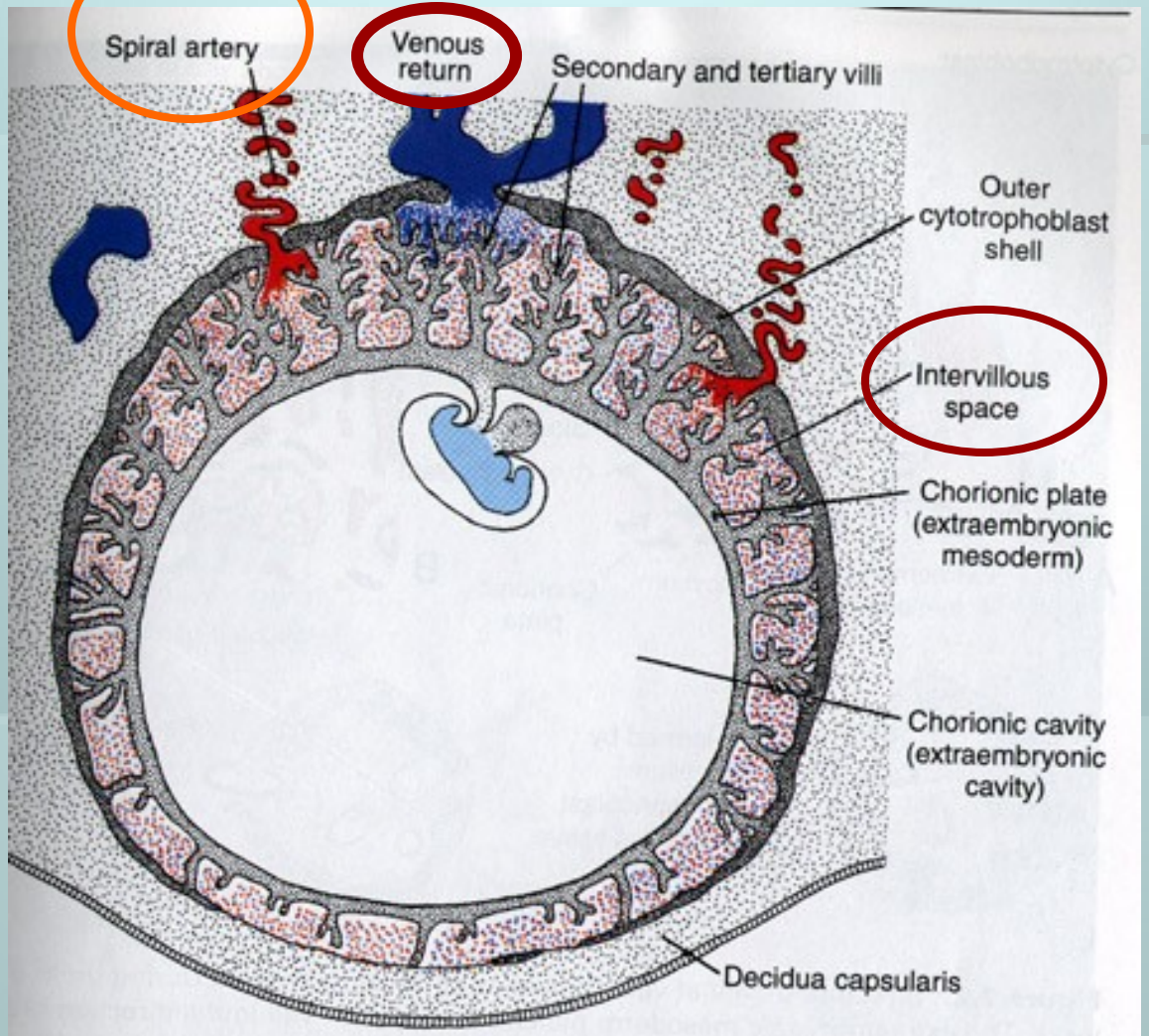
DEVELOPMENT OF PLACENTA

1. **MATERNAL PART_ Decidua Basalis/Basal P.**
2. **Caverous Intervillous Space**
3. **FETAL PART_ Chorionic villi/Chorionic P.**



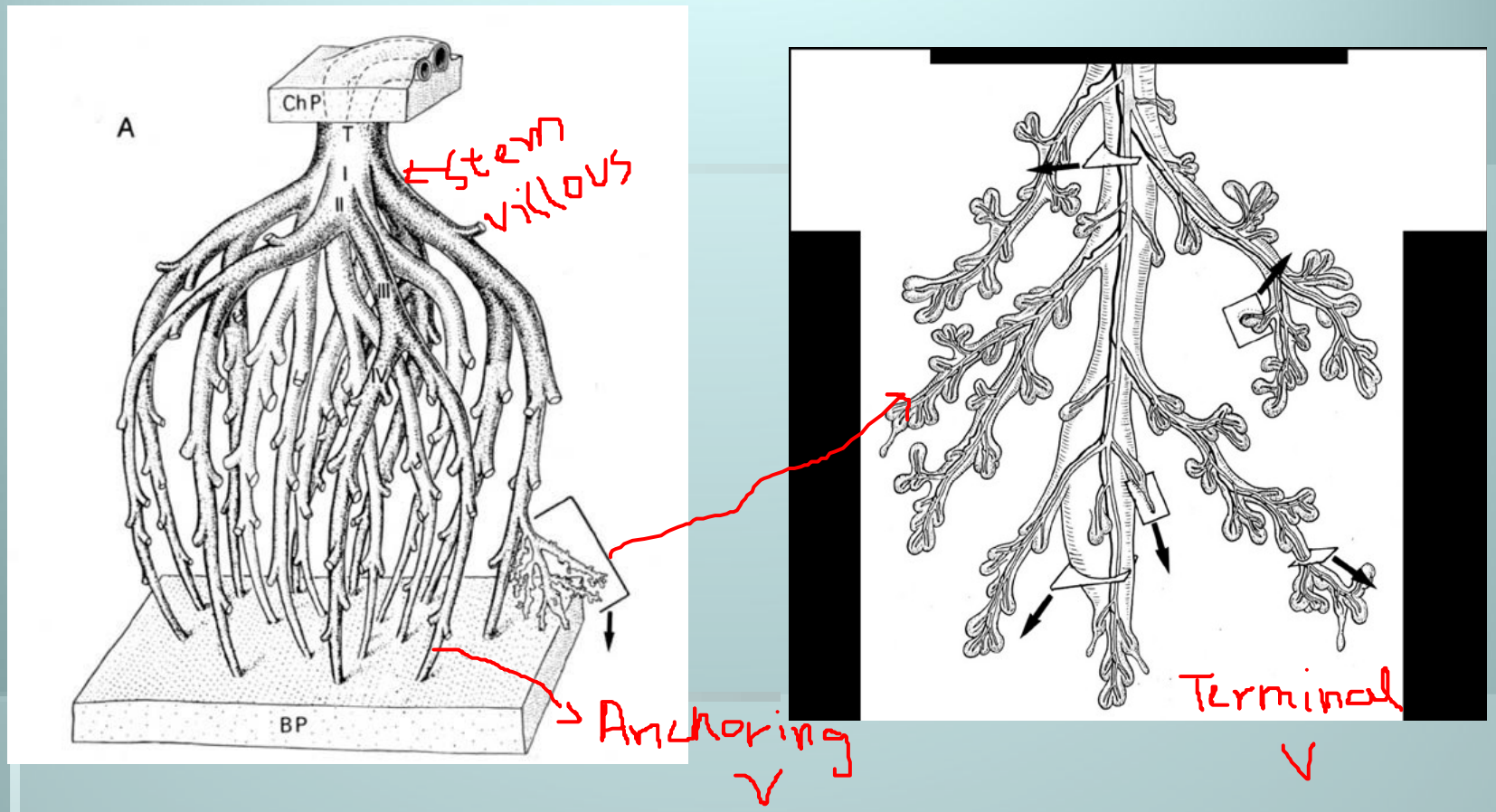
4/5th of placenta is of fetal origin and 1/5 is of maternal origin

VILLUS FORMATION



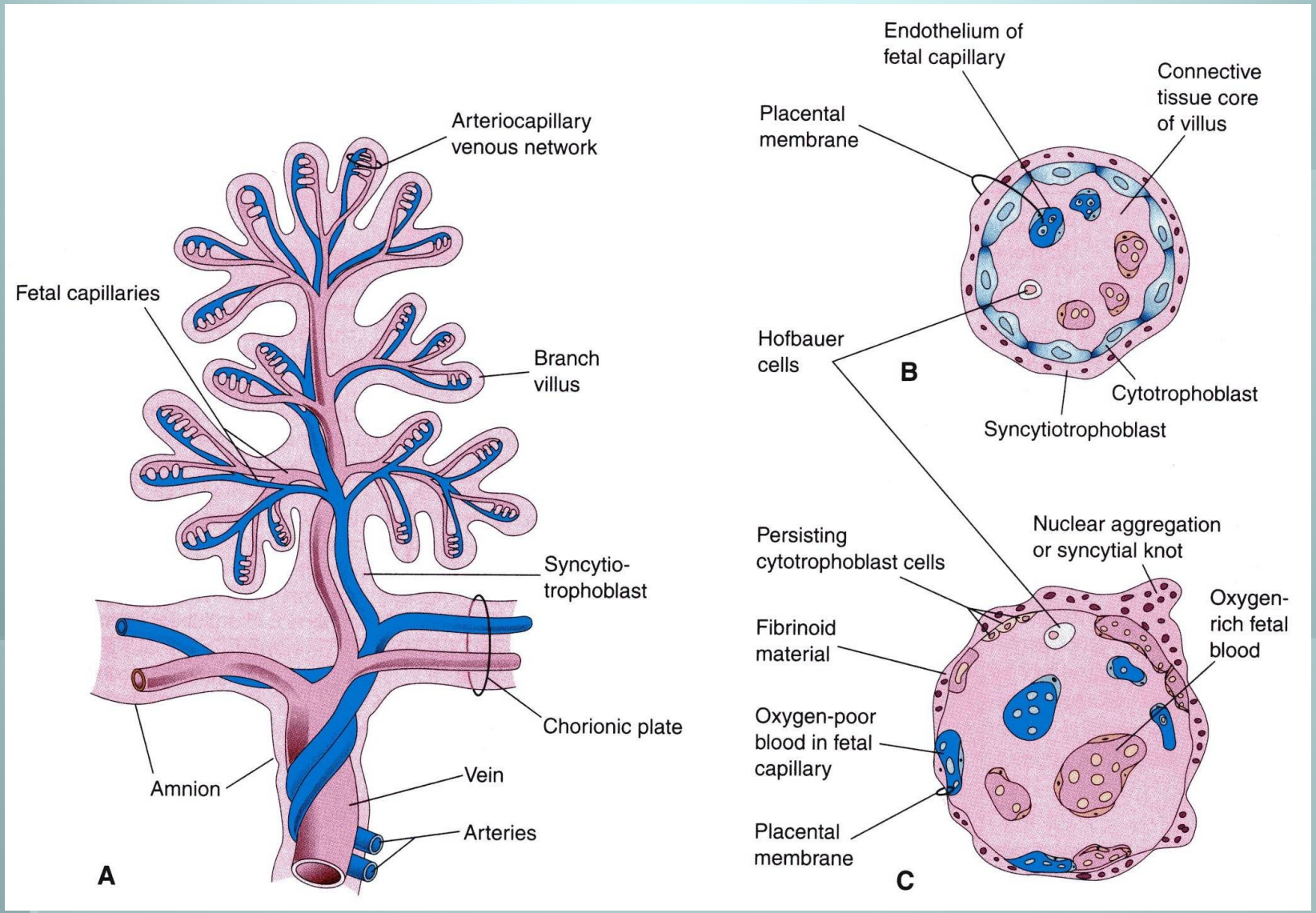
gure 7.1. Human embryo at the beginning of the second month of development. the embryonic pole, villi are numerous and well formed; at the abembryonic pole y are few in number and poorly developed.

VILLUS FORMATION

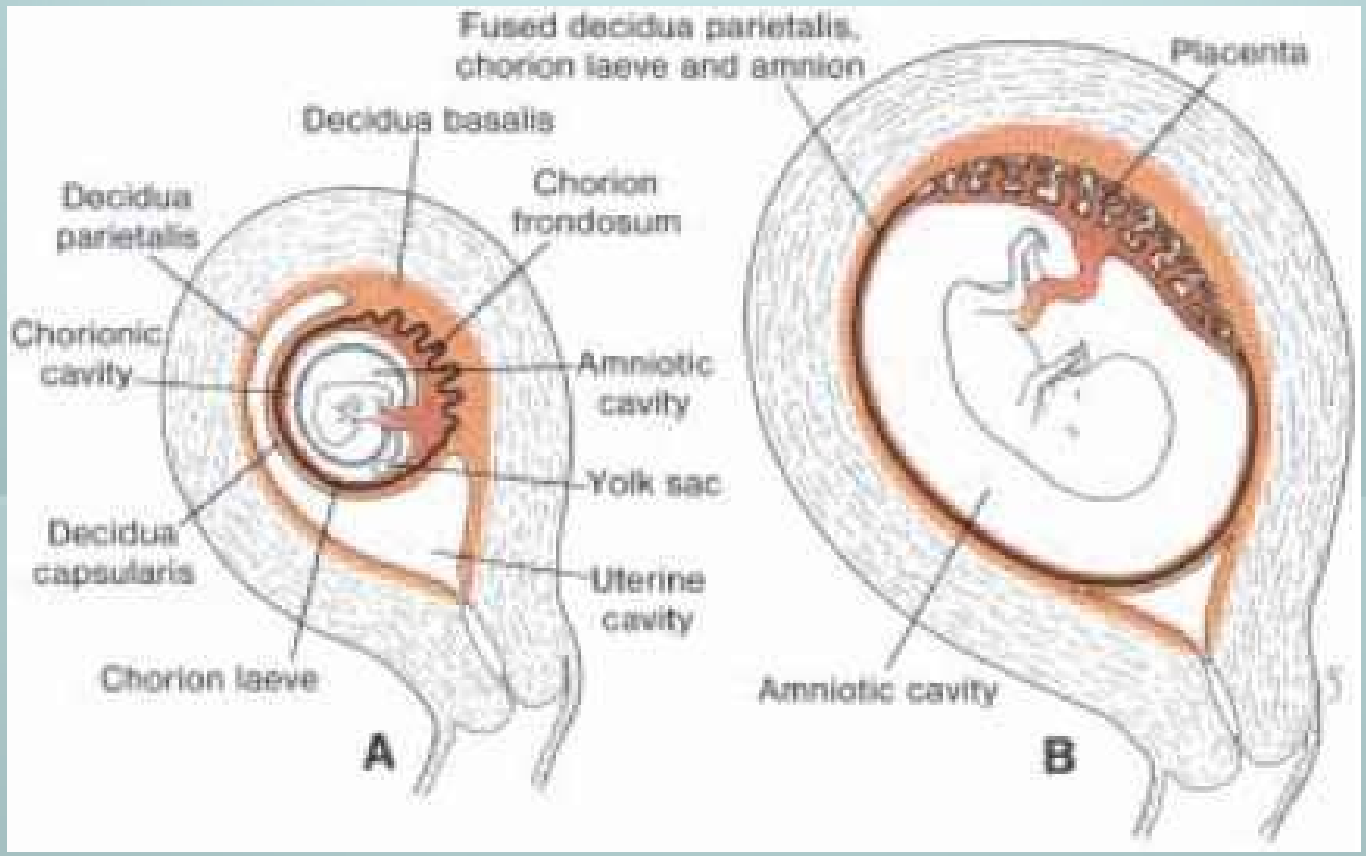


- Proximal part of CV becomes main stem villus (60) attaching to chorionic plate
- More distal parts become anchoring villi attached to basal plate

STRUCTURE OF STEM CHORIONIC VILLUS



Chorionic frondosum ready



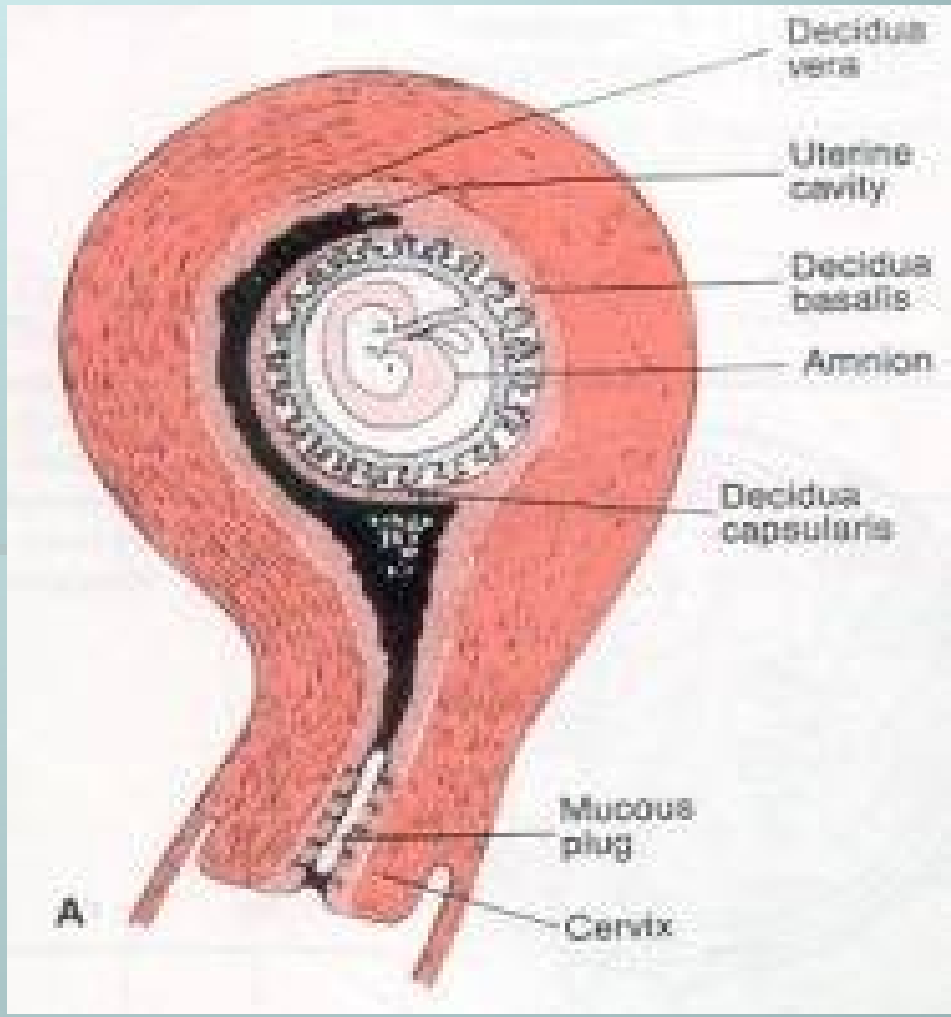
DEVELOPMENT OF PLACENTA

MATERNAL PART
DECIDUA

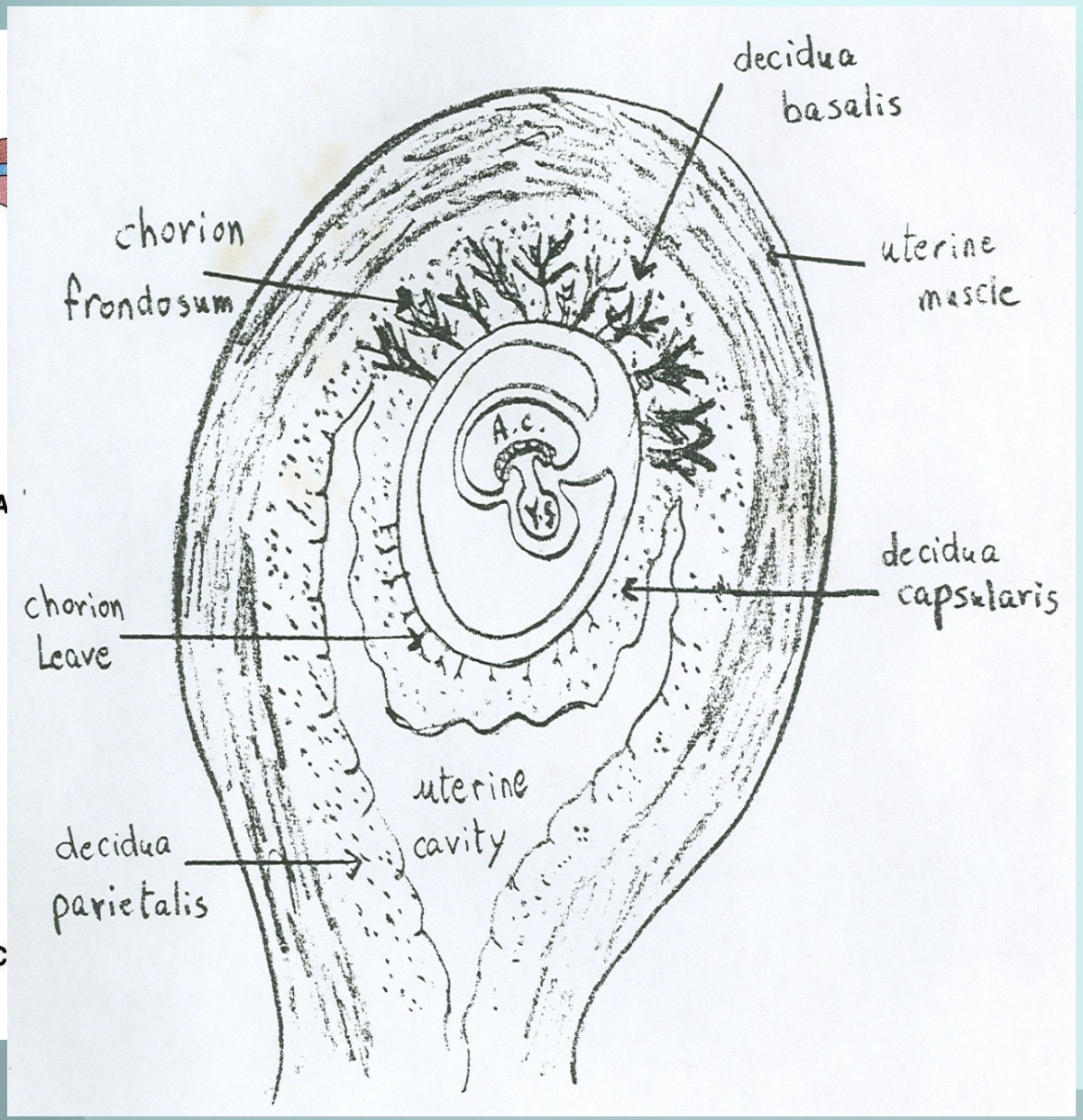
1. Functional layer of endometrium of gravid (pregnant) uterus

2. Includes endometrium of fundus & body of uterus

3. Endometrium of cervix does not form a part of decidua



DECIDUA



- **Decidua basalis** Lies at site of implantation ,it forms maternal part of placenta
- **Decidua capsularis**: it covers conceptus
- **Decidua parietalis:** rest of endometrium that lines body & fundus

DEVELOPMENT OF PLACENTA

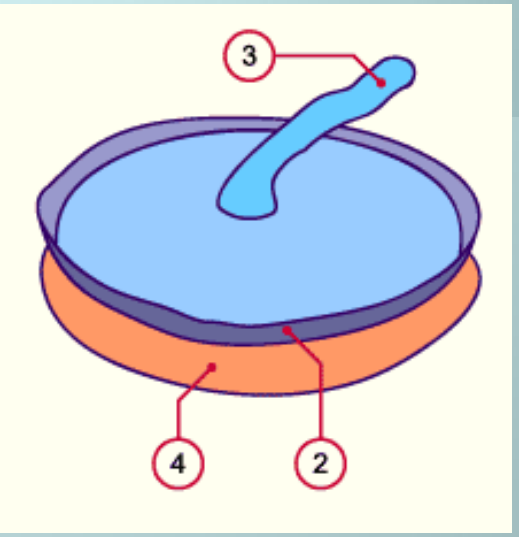
MATERNAL PART
DECIDUA

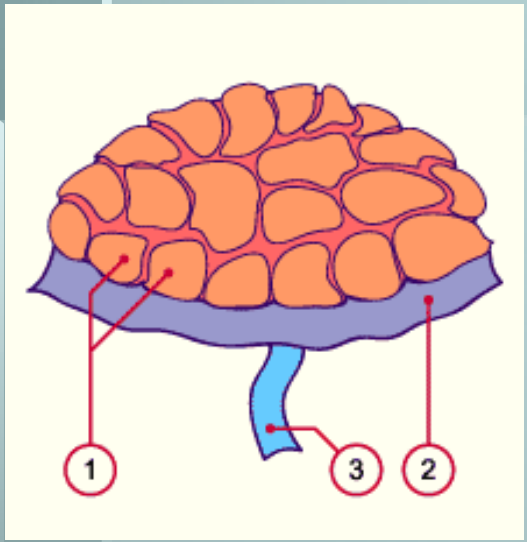
- **CAUSE OF DECIDUAL REACTION**

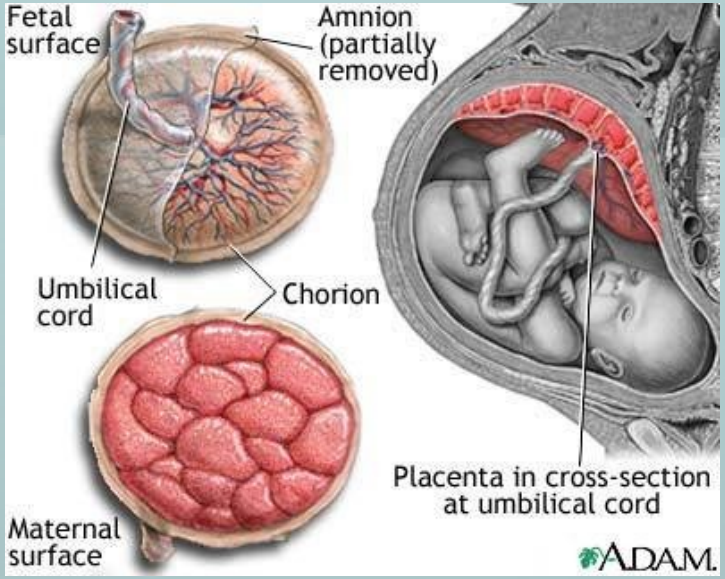
1. Trophoblast secretes chorionic gonadotrophins which prolong life of corpus luteum
2. Corpus luteum of pregnancy continues to secrete progesterone till 20th week
3. Increasing level of progesterone in maternal blood induces changes in endometrium into decidua

SURFACES OF PLACENTA

1. Fetal Surface
2. Maternal Surface







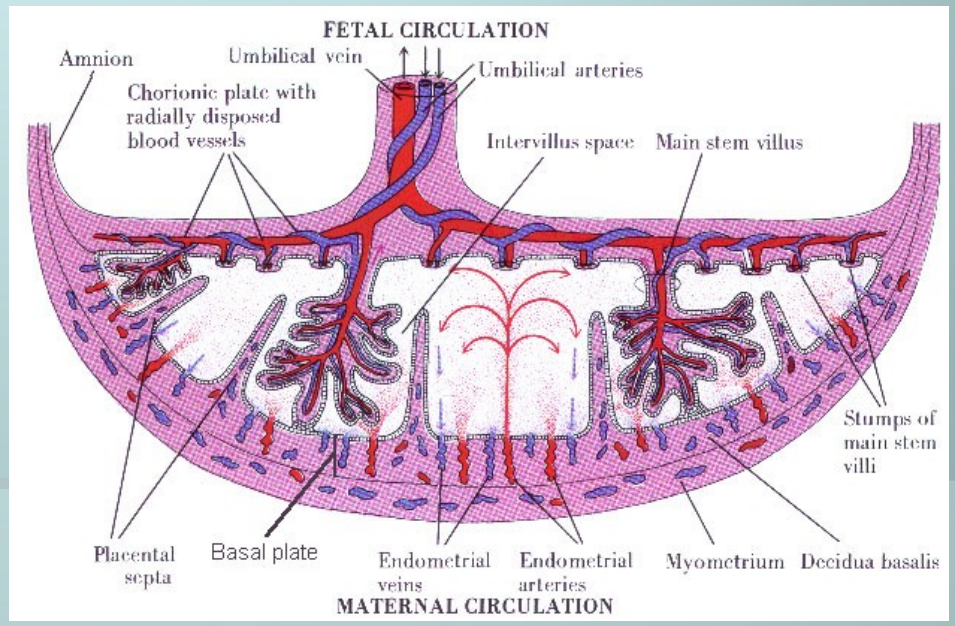


STRUCTURE OF PLACENTA

- **Amniotic membrane-** single layer of cuboidal epithelium loosely attached to adjacent chorionic plate and does not take part in placental formation

- **Chorionic plate-** forms roof of placenta
- From outside inwards consists of

1. Syncitotrophoblast
2. Cytotrophoblast
3. Extraembryonic mesoderm with branches of umbilical vessels

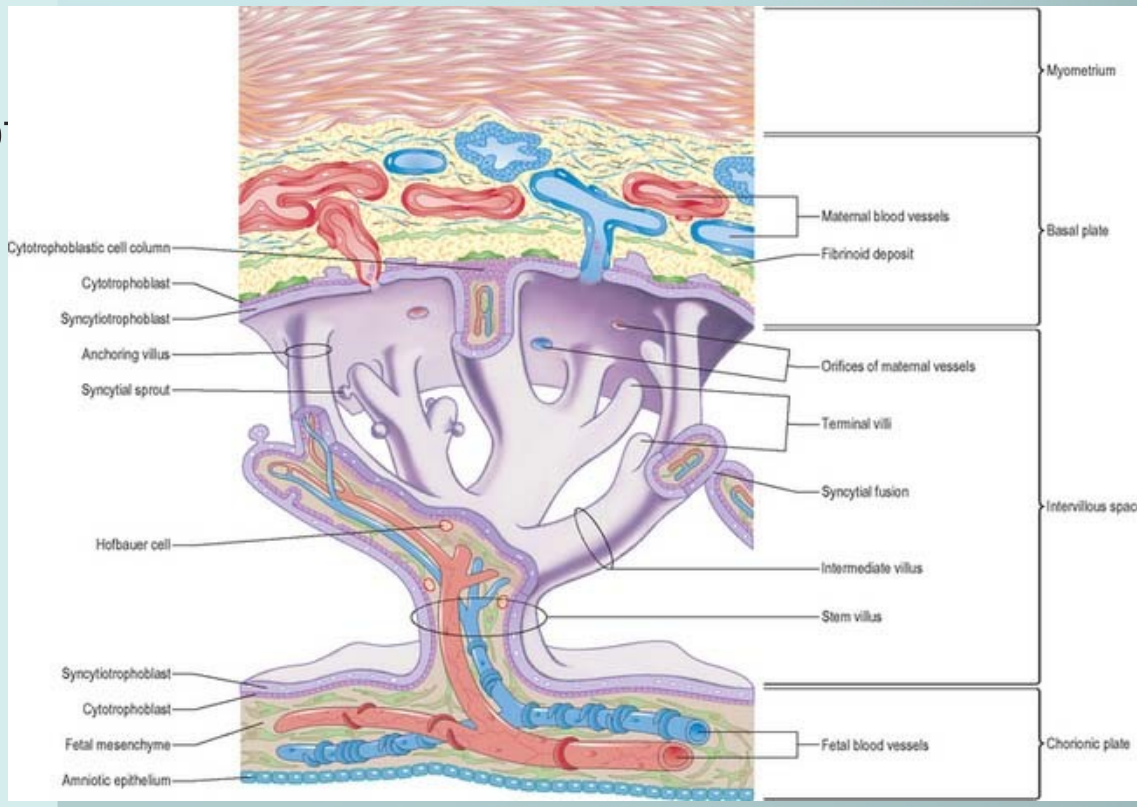


STRUCTURE OF PLACENTA

- **Basal Plate-** forms floor From outside inwards:

  - Compact and spongy layer of decidua basalis
  - Layer of Nitabuch
  - Cytotrophoblastic shell
  - Syncytiotrophoblast

Basal plate is perforated by spiral arteries allowing entry of maternal blood into intervillous space



STRUCTURE OF PLACENTA

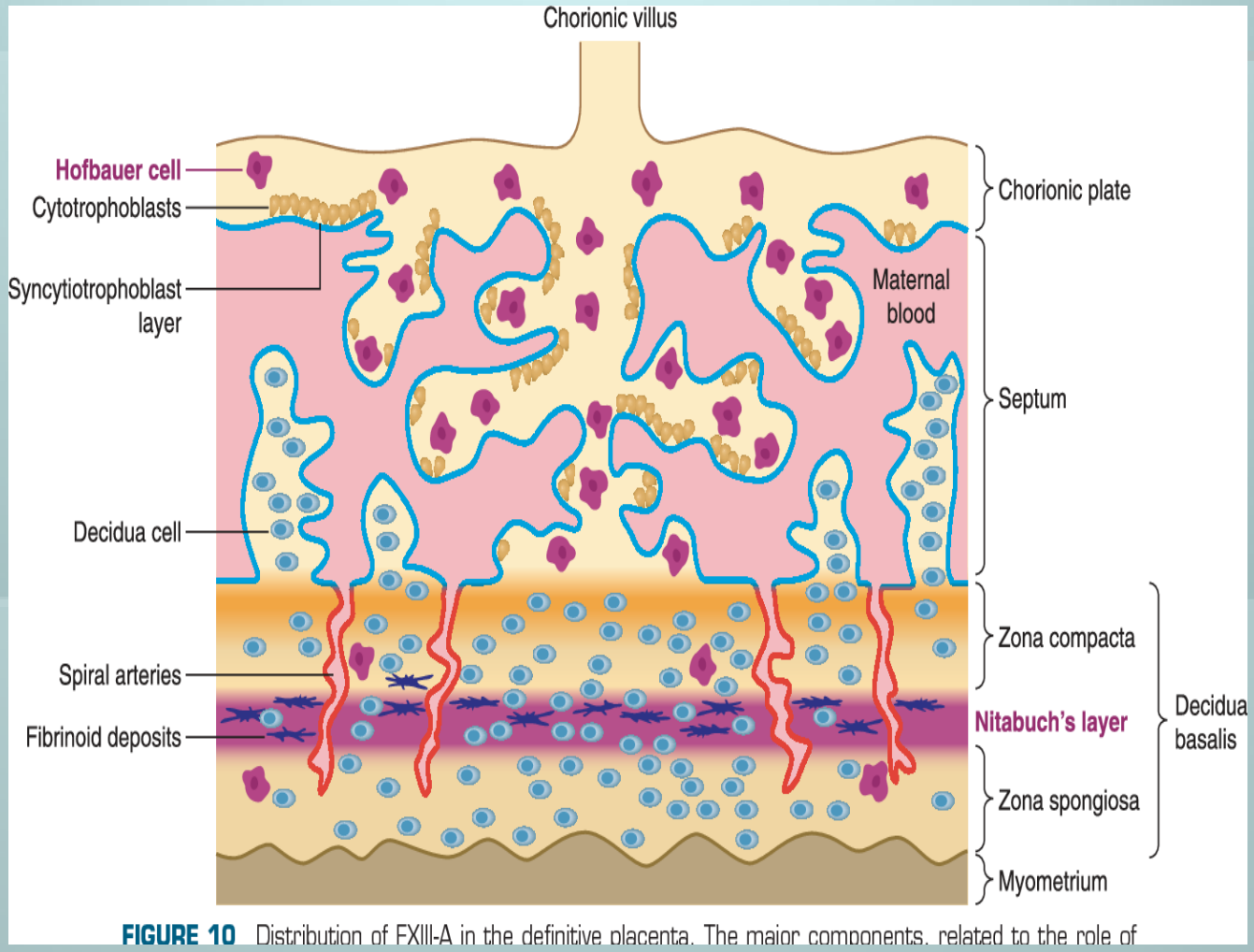
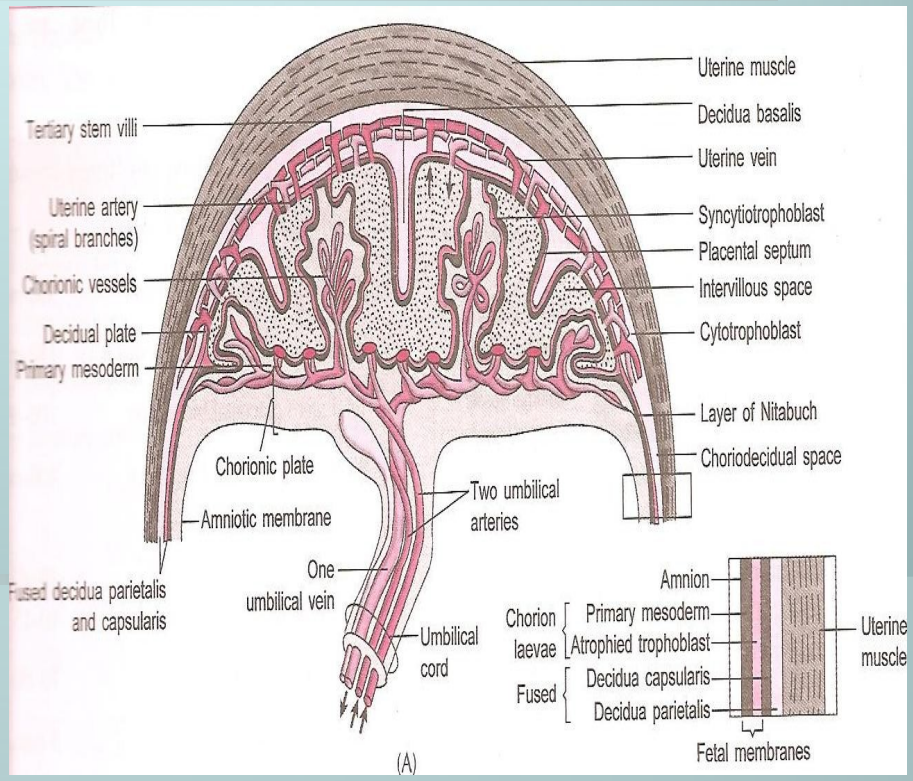


FIGURE 10 Distribution of FXIII-A in the definitive placenta. The major components, related to the role of

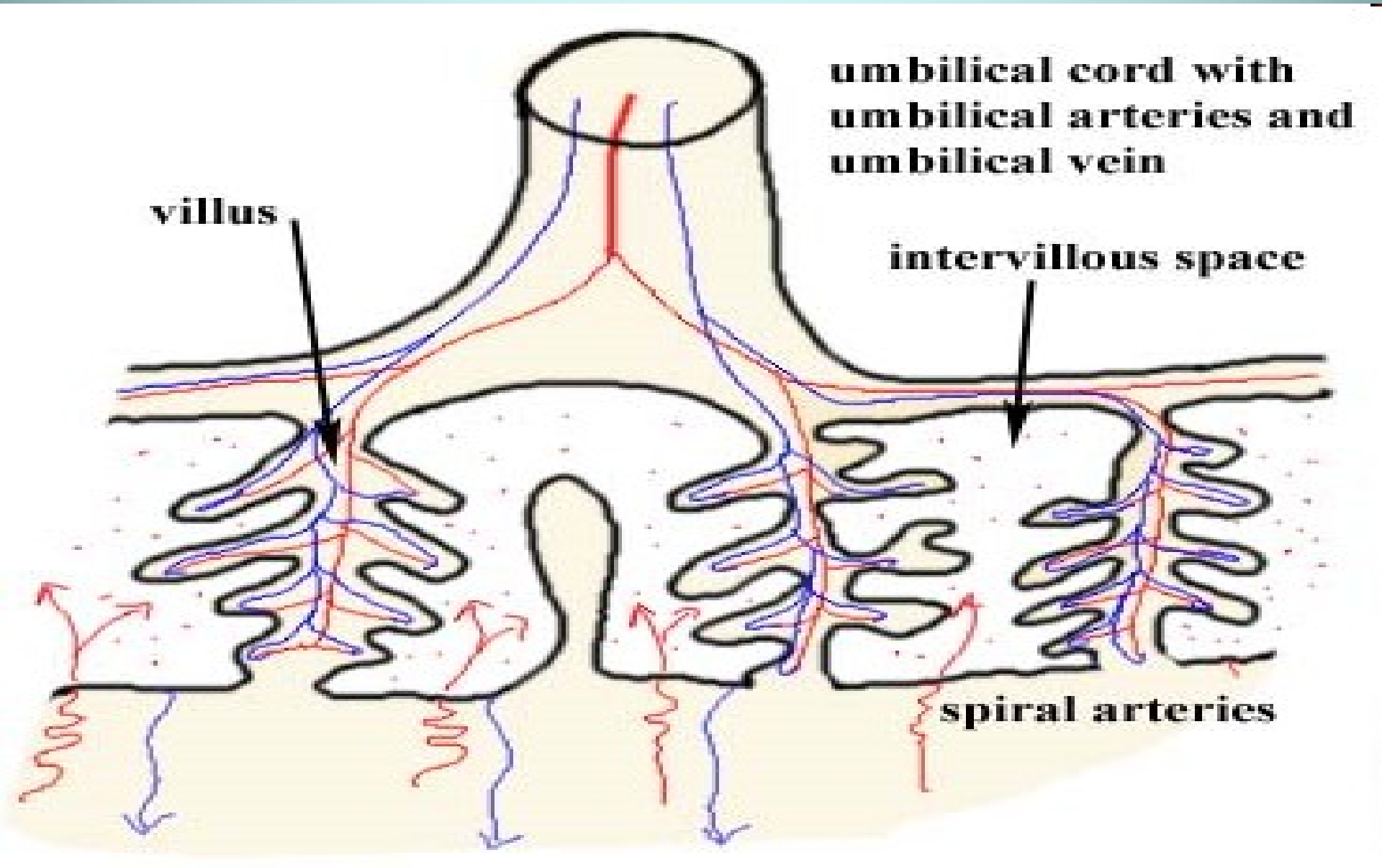
STRUCTURE OF PLACENTA

Intervillous space

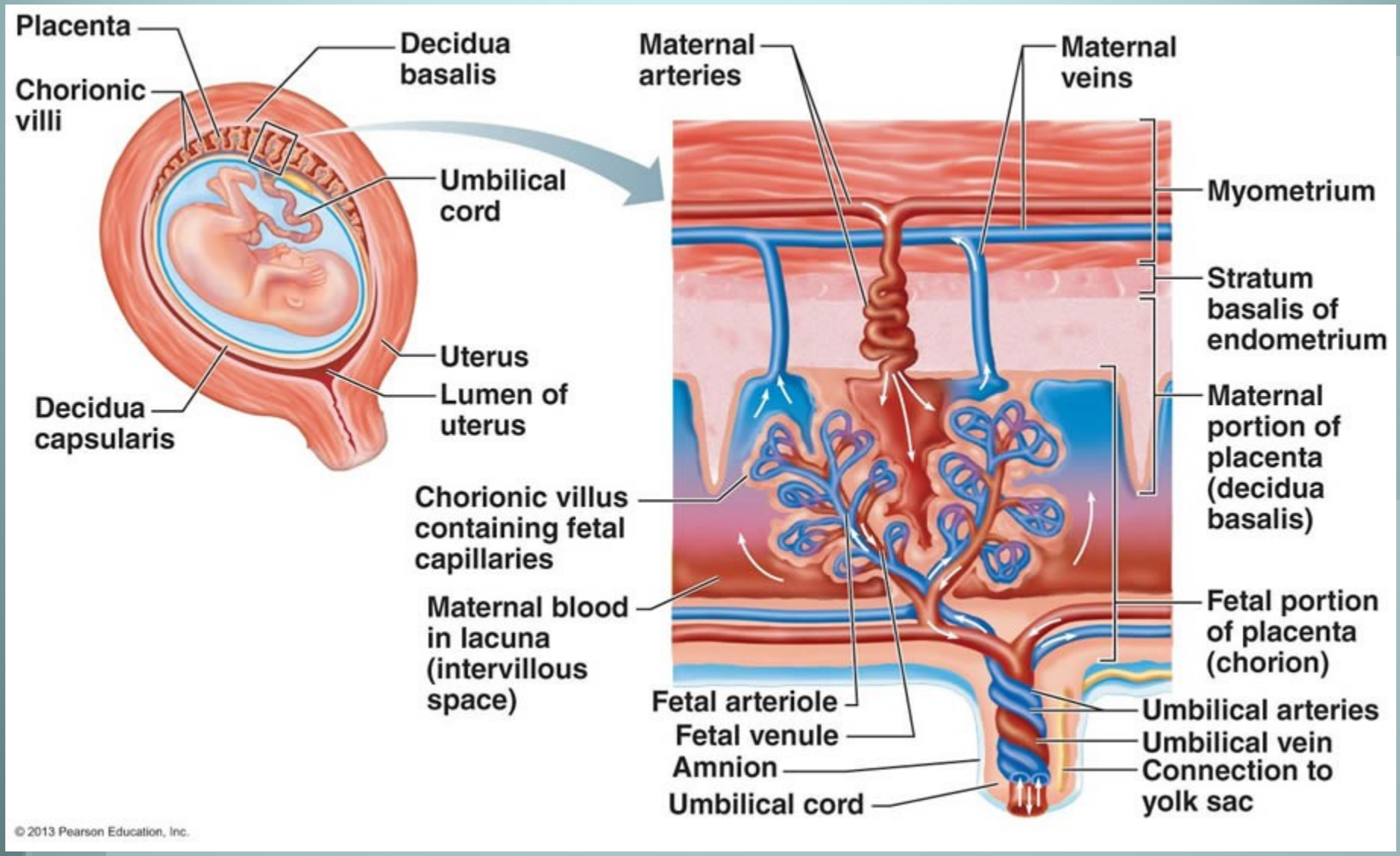
- Numerous branch villi arising from stem villi project into this space
- Lined internally on all sides by syncytiotrophoblast and is filled with maternal blood



PLACENTAL CIRCULATION



PLACENTAL CIRCULATION

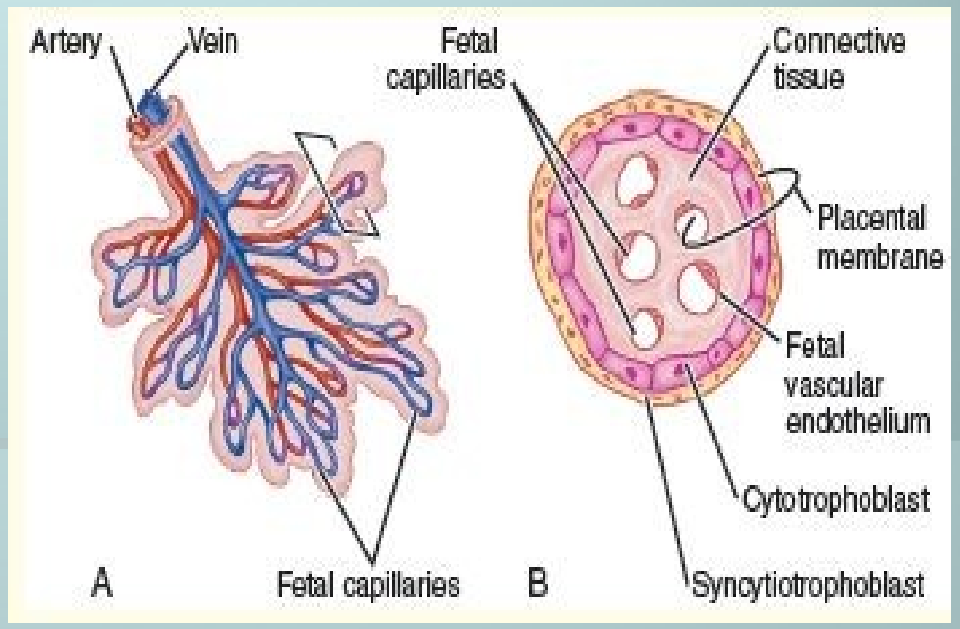


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Placental barrier or membrane

- Maternal and fetal blood are separated by placental membrane or barrier

1. **Endothelial lining of fetal vessels**
2. **Basement membrane**
3. **Connective tissue of villi**
4. **Cytotrophoblast**
5. **Syncytiotrophoblast**



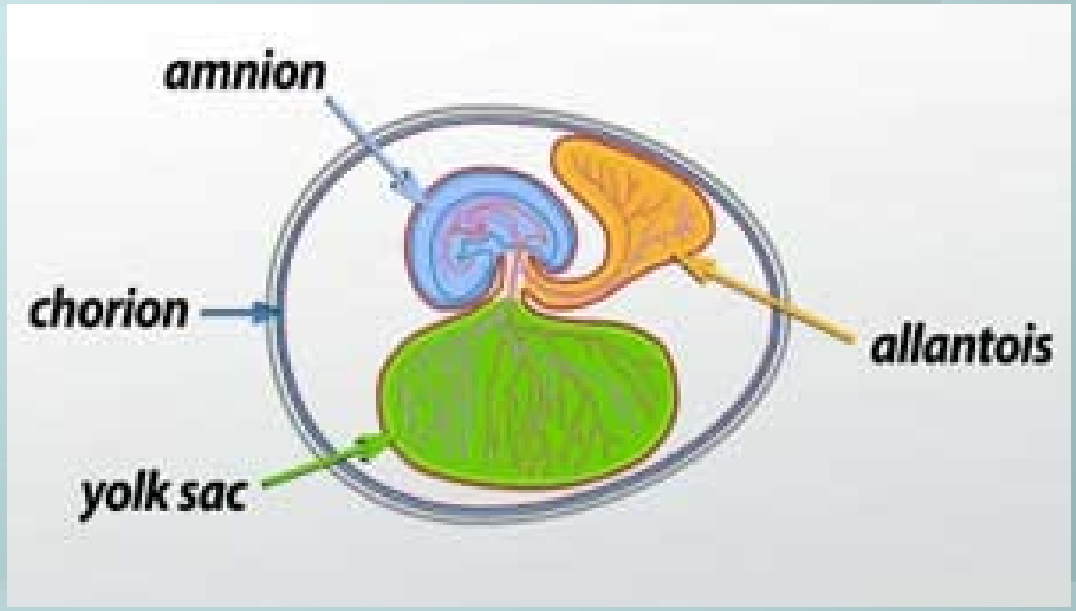
Placental barrier or membrane

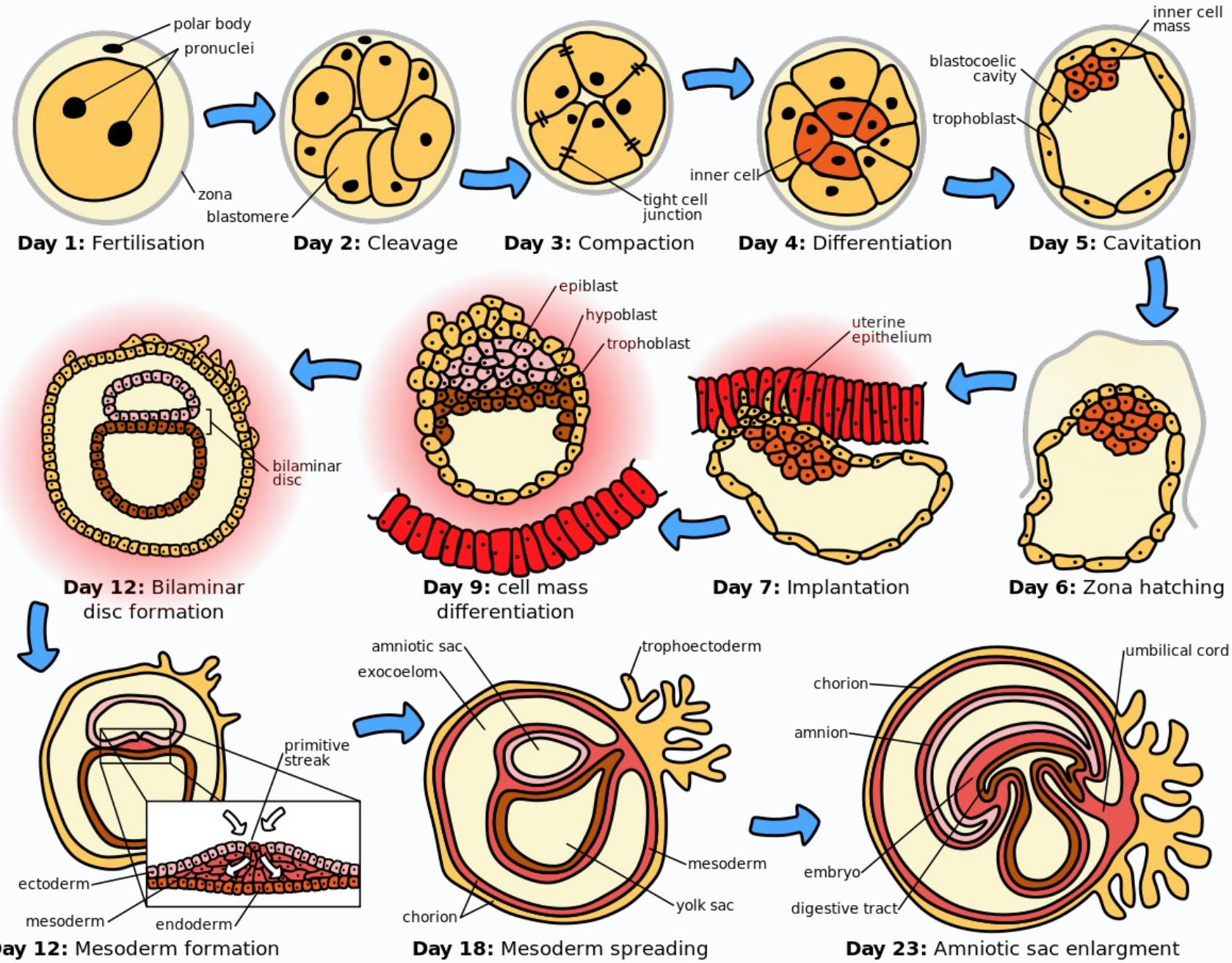
- Separates fetal from maternal blood
- Prevents mixing of them
- Is an incomplete (semi permeable) barrier as it only prevents large molecules to pass (heparin & bacteria)
- **Rh-positive fetal blood** can enter circulation of an Rh-negative mother. The mother will make an antibody which then passes through placenta to fetus causing **haemolytic disease**
- Cannot prevents passage of viruses(e.g. rubella), micro-organisms(toxoplama, treponema pallidum) drugs and hormones

Fetal membrane — overview

- Origin

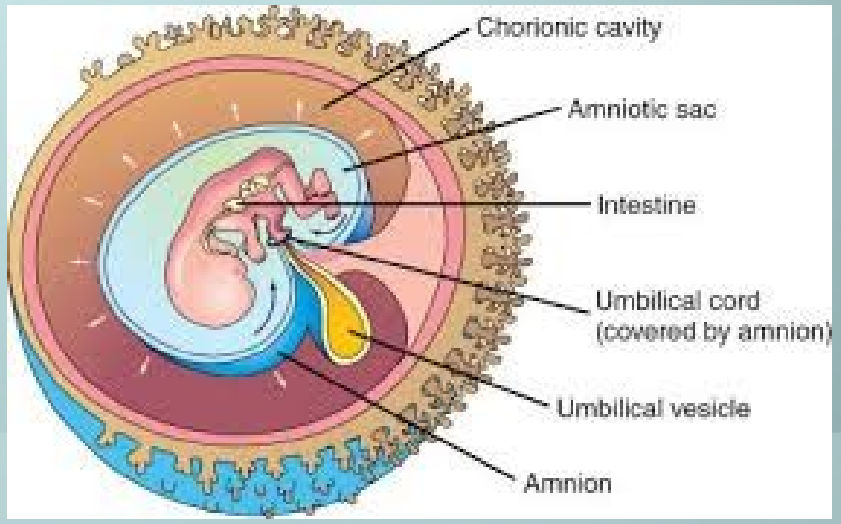
1. Chorion
2. Amnion
3. Yolk sac
4. Allantois
5. Umbilical cord




polar body
pronuclei
zona
blastomere
Day 1: Fertilisation
Day 2: Cleavage
tight cell junction
Day 3: Compaction
inner cell
Day 4: Differentiation
inner cell mass
blastocoelic cavity
trophoblast
Day 5: Cavitation
Day 6: Zona hatching
uterine epithelium
Day 7: Implantation
epiblast
hypoblast
trophoblast
Day 9: cell mass differentiation
bilaminar disc
Day 12: Bilaminar disc formation
primitive streak
ectoderm
mesoderm
endoderm
Day 12: Mesoderm formation
amniotic sac
exocoelom
trophoectoderm
mesoderm
yolk sac
chorion
Day 18: Mesoderm spreading
chorion
amnion
umbilical cord
embryo
digestive tract
Day 23: Amniotic sac enlargment

AMNION

- Inner membrane building amniotic sac that surrounds and protects an embryo
- Thin but tough
- Maximum tensile strength
- Forms a fluid filled membranous amniotic sac that surrounds embryo and fetus
- Is attached to margins of embryonic disc
- Its junction with embryo located on the ventral surface after folding

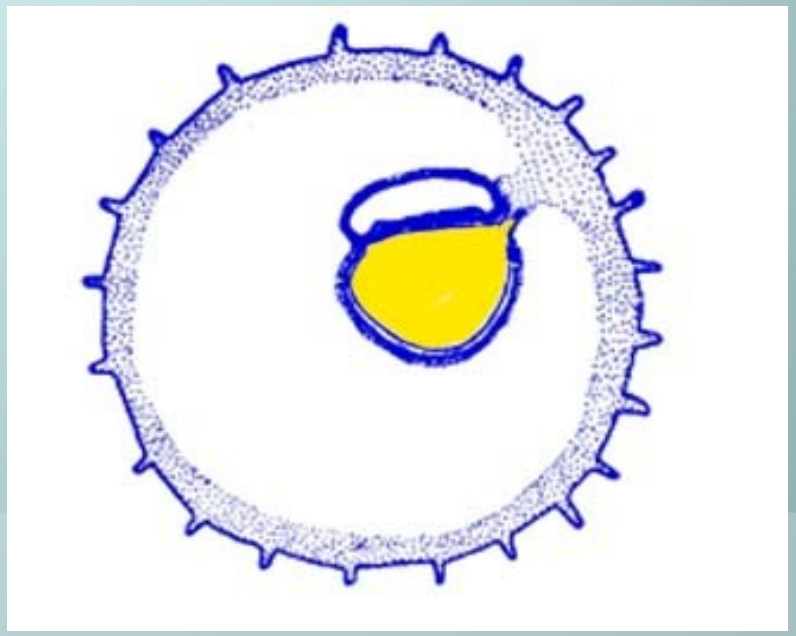


FORMATION OF AMNION

- Develops at end of day 7

Boundaries

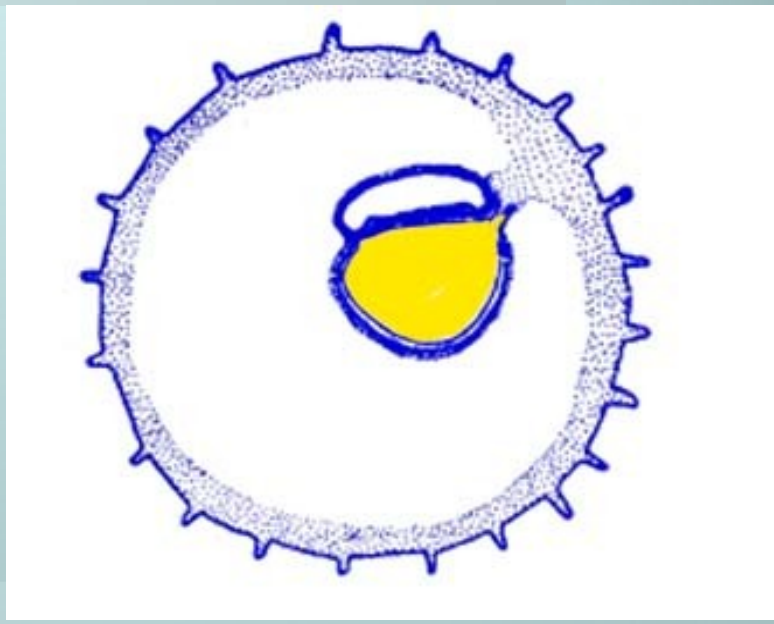
- Roof is made up of a single layered flat cells ectoderm
- Floor consists of ectoderm of embryonic disc
- Outside ectoderm is a thin layer of mesoderm, continuous with that of somatopleure and is connected by body-stalk with mesodermal lining of chorion



FORMATION OF AMNION

AMNIOTIC FLUID

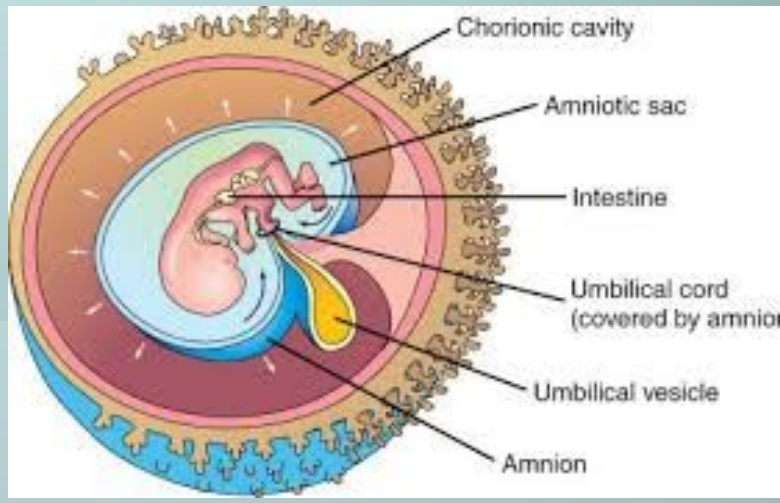
- ***Liquid contained by amniotic sac***
- At first amnion is in contact with embryo
- By 4th week, fluid secreted by aminoblast begin to accumulate
- Fluid increases in quantity till 6 and 7 months and causes amnion to expand and ultimately to adhere to inner surface of chorion, so that extra-embryonic part of the celom is obliterated



At the end of pregnancy it is about 1 liter

AMNIOTIC FLUID
FUNCTIONS

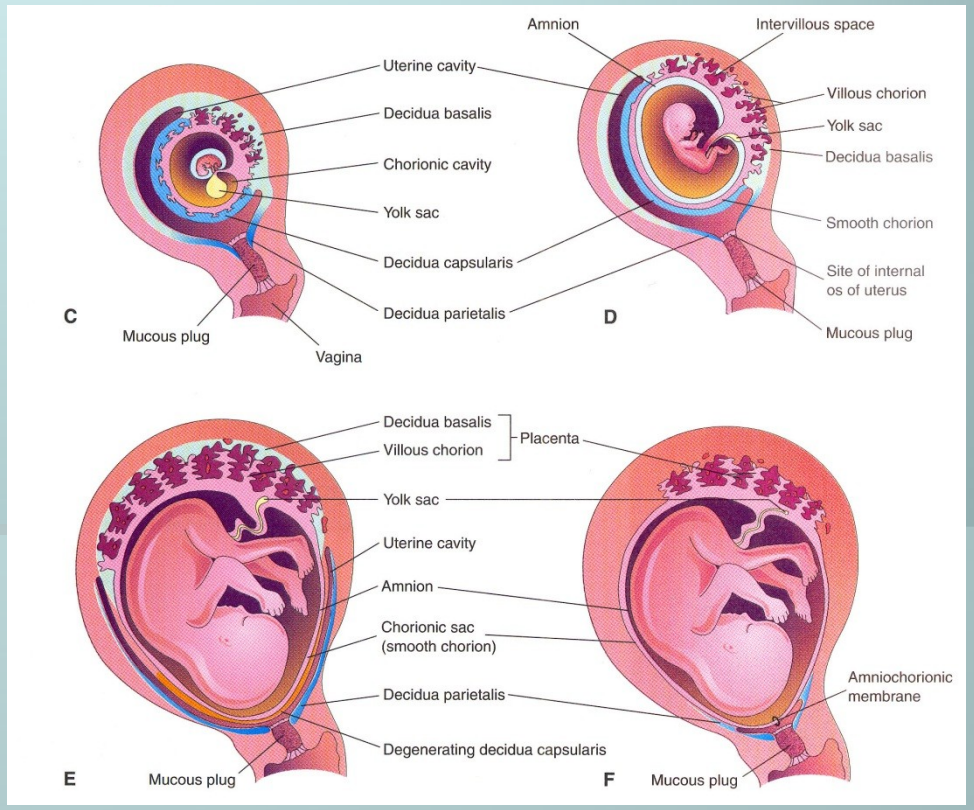
- Inhaled and exhaled by fetus
- Essential that fluid be breathed into lungs in order for them to develop normally
- Swallowed amniotic fluid also creates urine and contributes to formation of meconium
- Mechanical cushion
- Protect from fetus adhesion
- Fetal Movement
- Maintain Temp: Helps protect fetus from heat loss



AMNIOTIC FLUID
Abnormalities

1) too much (polyhydramnios)
   - >2000 ml
   - Abnormal digestive system or CNS
   - - esophageal atresia
   - - anencephaly

2) too little (oligohydramnios)
   - <500 ml
   - Abnormal urinary system
   - - poor development of kidney
   - - urethra atresia



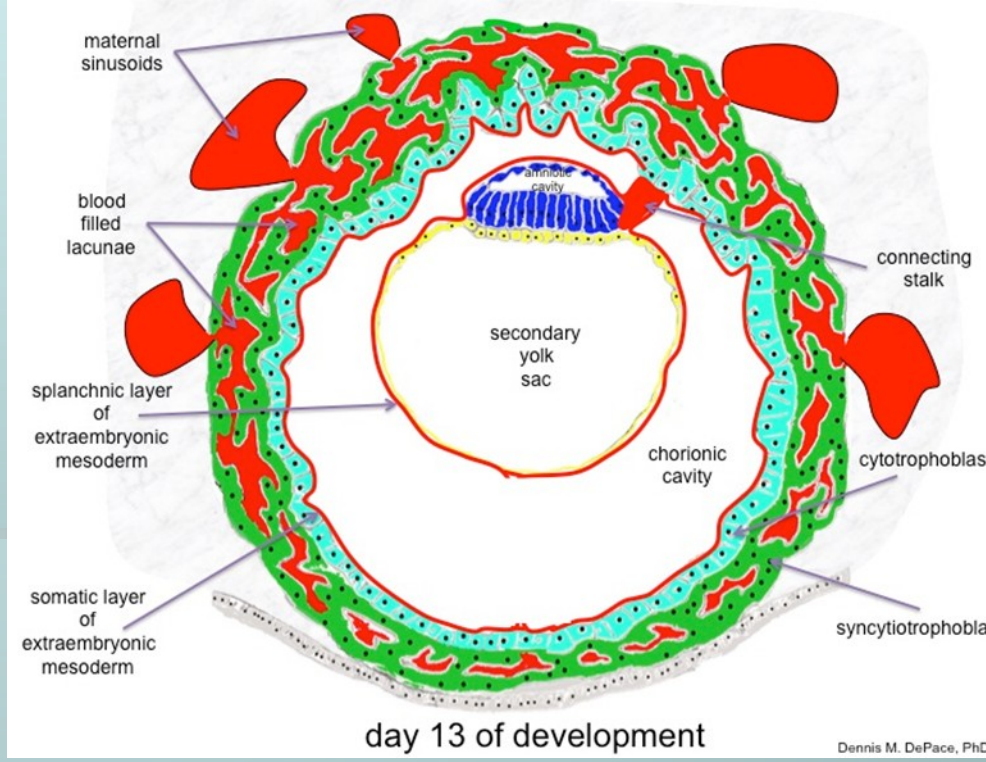
AMNIOCENTESIS

- Analysis of amniotic fluid, drawn out of mother's abdomen
- contains fetal cells, which can be examined for genetic defects
- Reveal many aspects of baby's genetic health



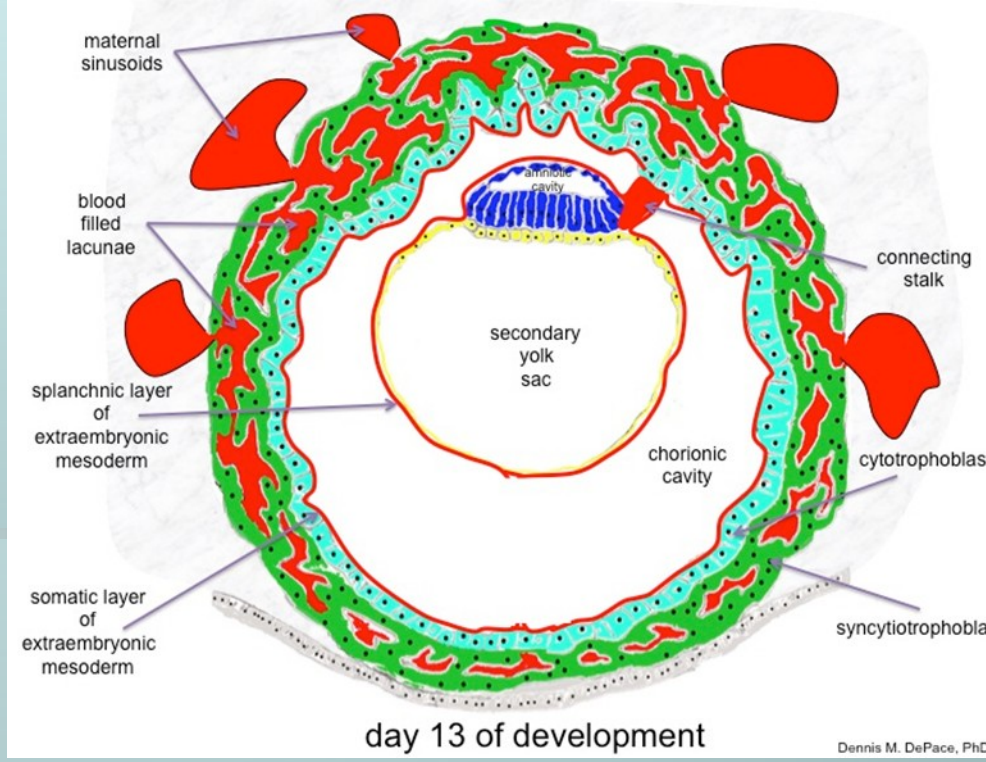
CHORION

The extraembryonic coelom expands and form a large cavity, the chorionic cavity



CHORION

- Extraembryonic somatic mesoderm and two layers of trophoblast form chorion
- Chorion forms wall of chorionic sac
- Embryo and its amniotic and yolk sacs are suspended into it by connecting stalk
- The extraembryonic coelom is now called chorionic cavity



CHORION

Formation

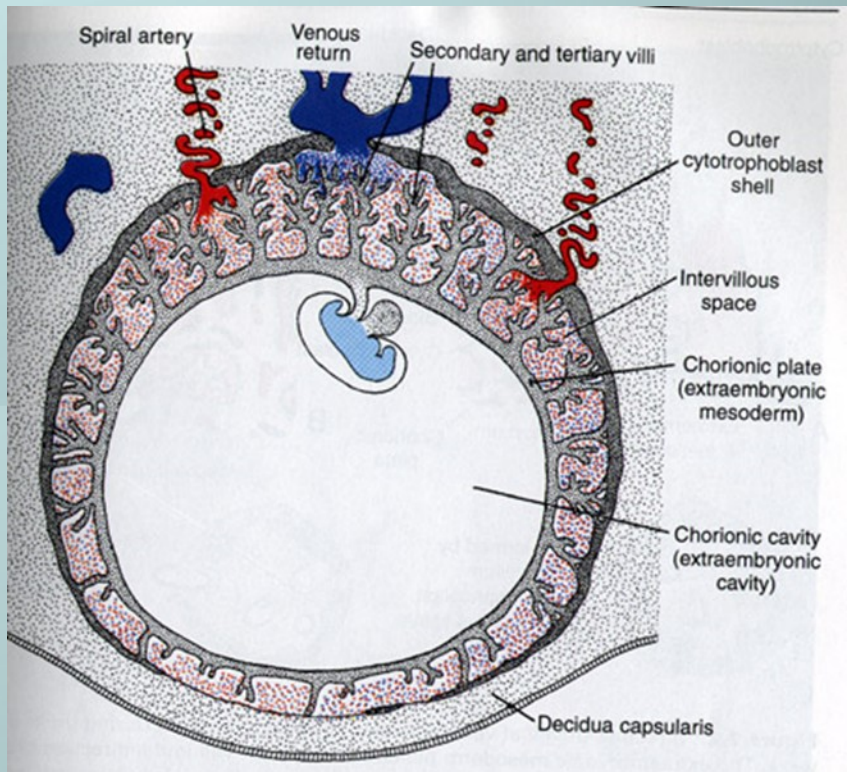
trophoblast +extraembryonic mesoderm

Chorion frondosum

(bushy chorion)- embryonic pole

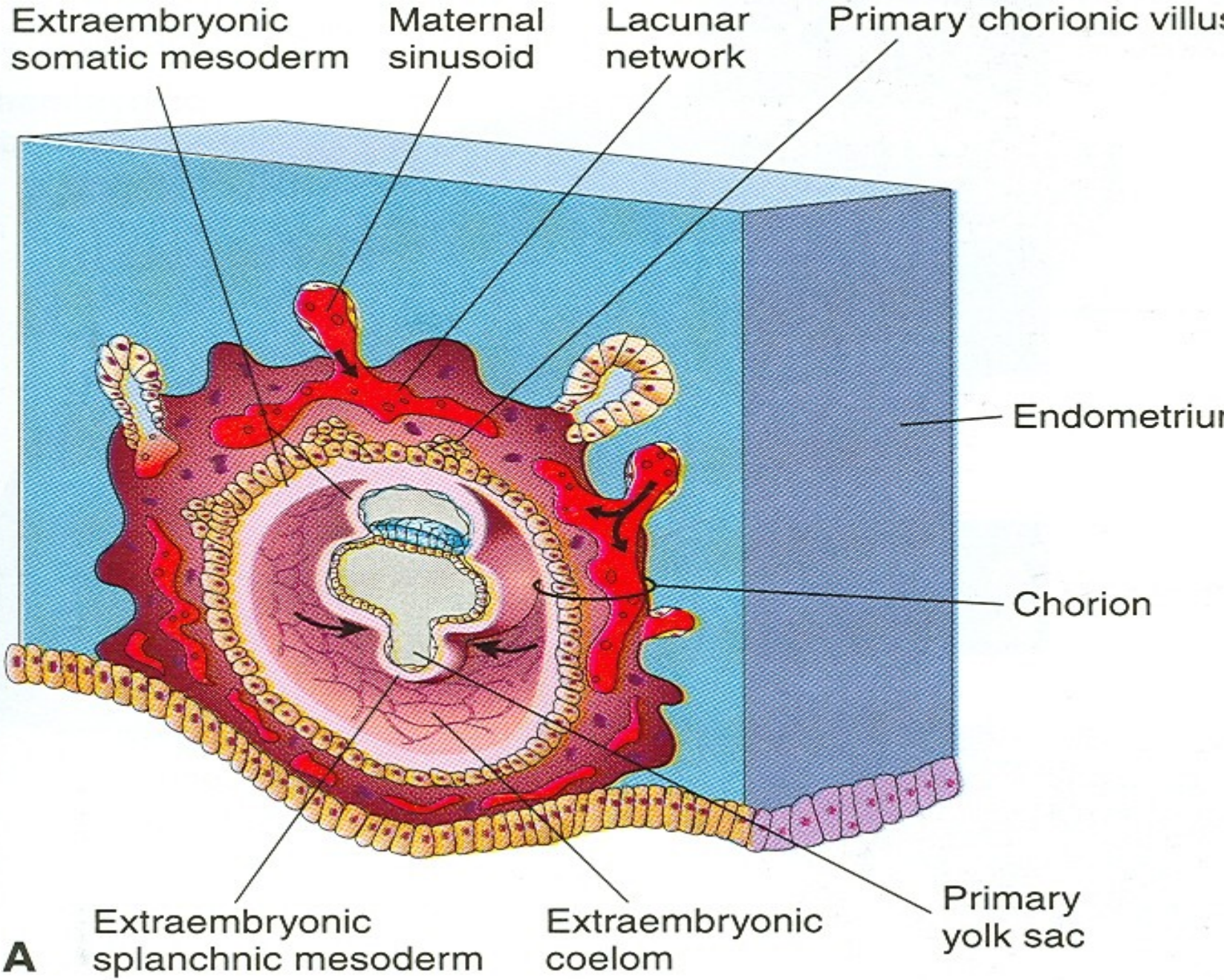
Chorion laeve

(smooth chorion)- abembryonic pole



ure 7.1. Human embryo at the beginning of the second month of development. the embryonic pole, villi are numerous and well formed; at the abembryonic pole y are few in number and poorly developed.

FETAL MEMBRANES



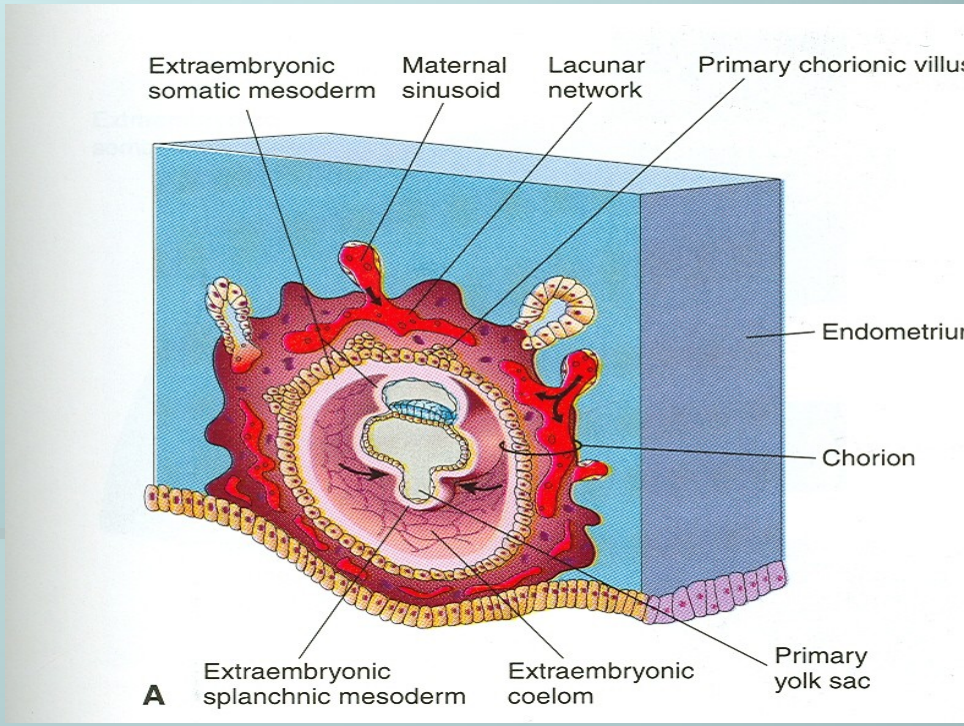
Function of Chorion

1) Exchange of metabolite

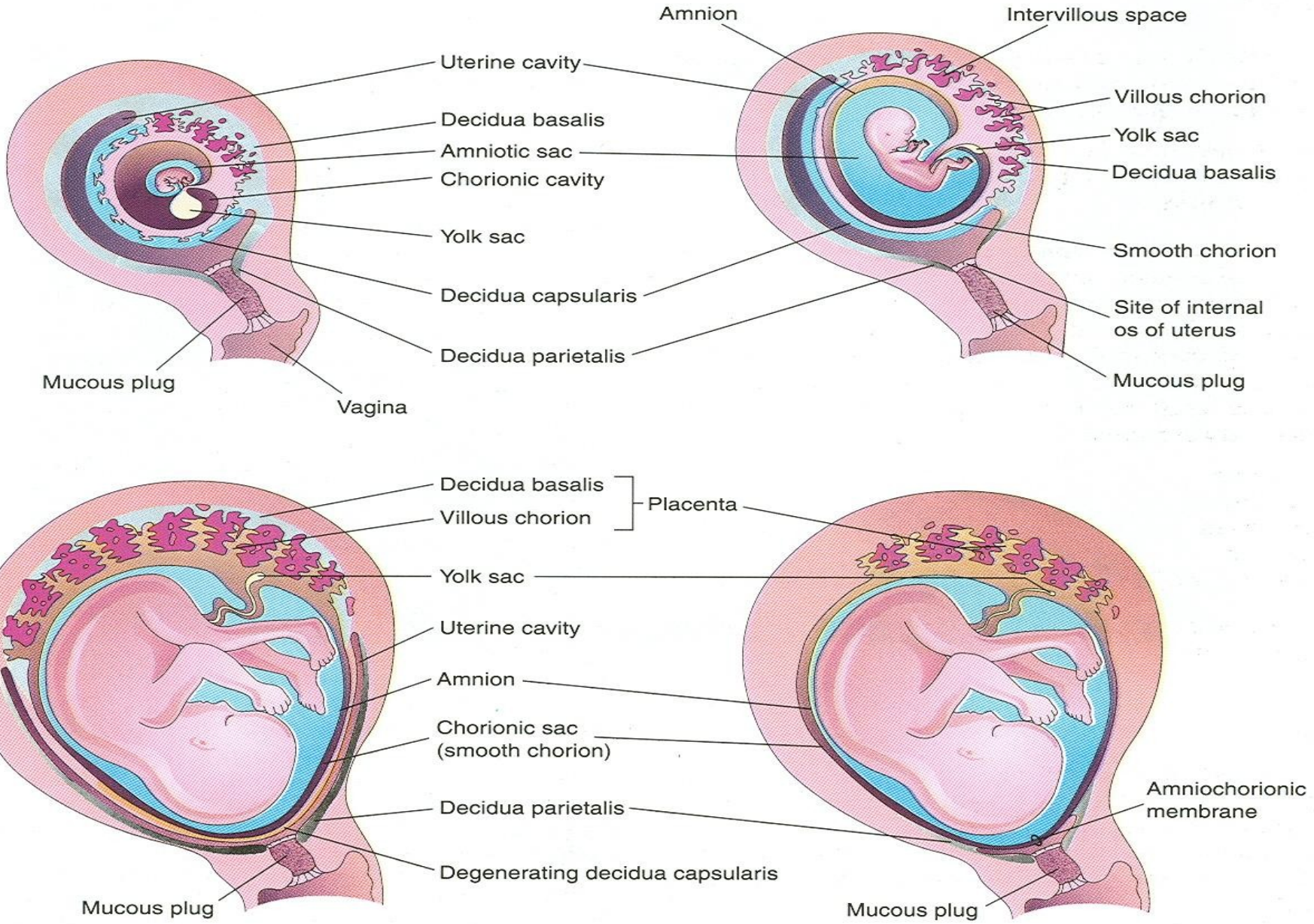
portion of placenta (Chorion frondosum)

2) Hormone production

human chorionic gonadotropin (HCG)

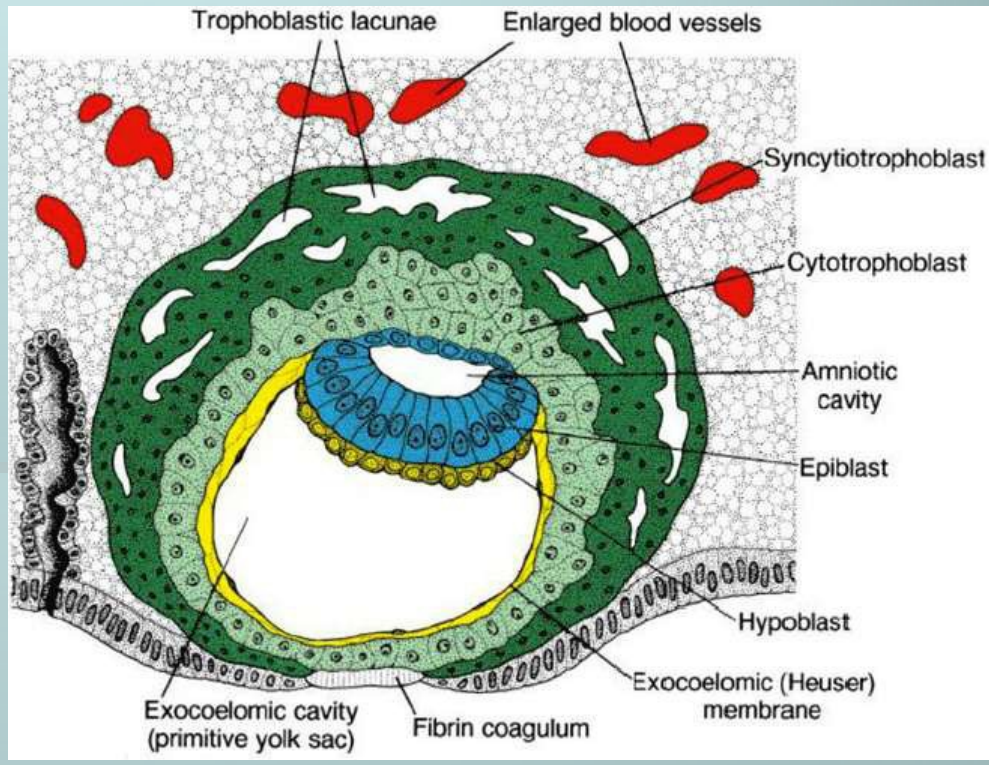


FETAL MEMBRANES



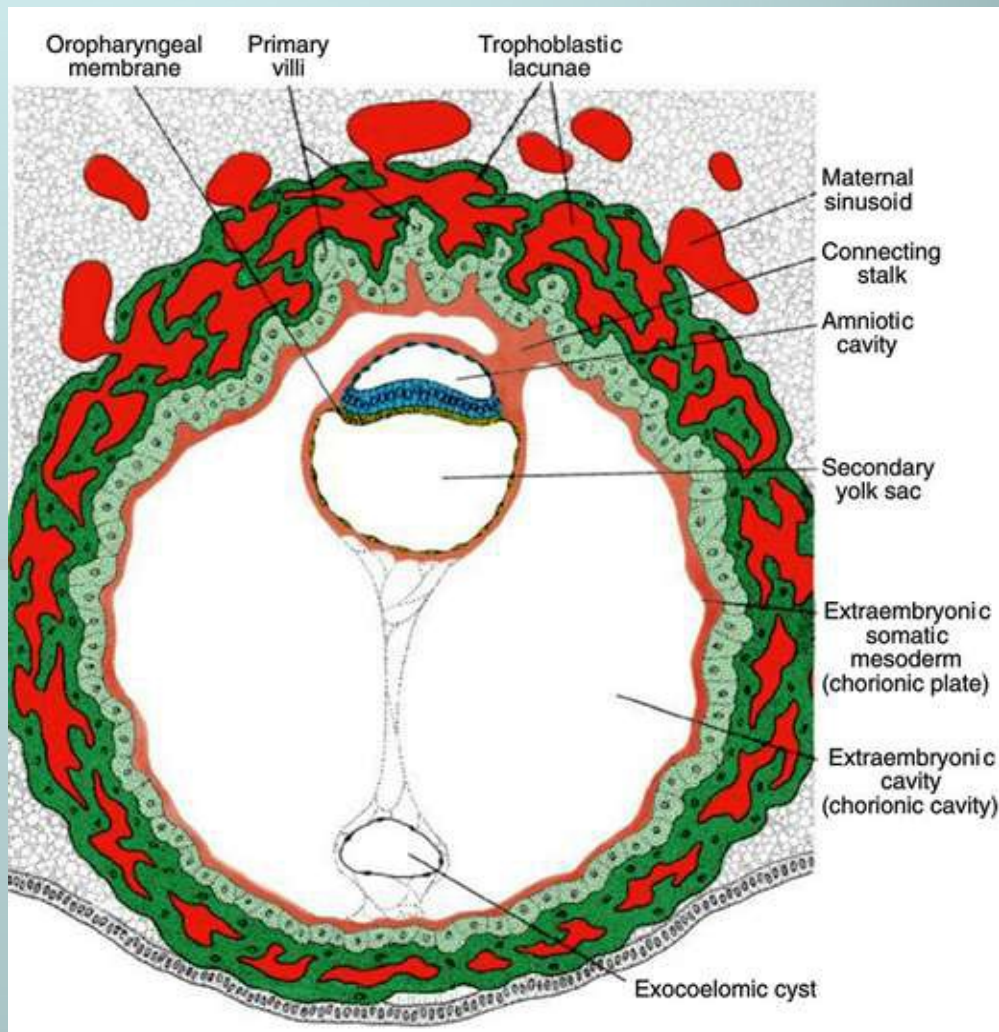
Yolk Sac (Umbilical vesicle)

- Shrinks to pear shaped remnant by 10th week & connected to midgut by a narrow yolk stalk
- Becomes very small at 20 weeks
- Usually not visible thereafter
- *Function*
  - Nutritive material is absorbed from yolk-sac and conveyed to embryo



Yolk Sac (Umbilical vesicle)

- 2 stages
- **Primary yolk sac** - has a wall composed of only of extraembryonic mesoderm cells that are organized in the exocoelomic membrane (of Heuser) - 2nd week
- **Secondary yolk sac** - has 2-layered wall lined by **endoderm** that has been proliferated along the inner surface of the exocoelomic membrane from the embryonic disc - on day 13

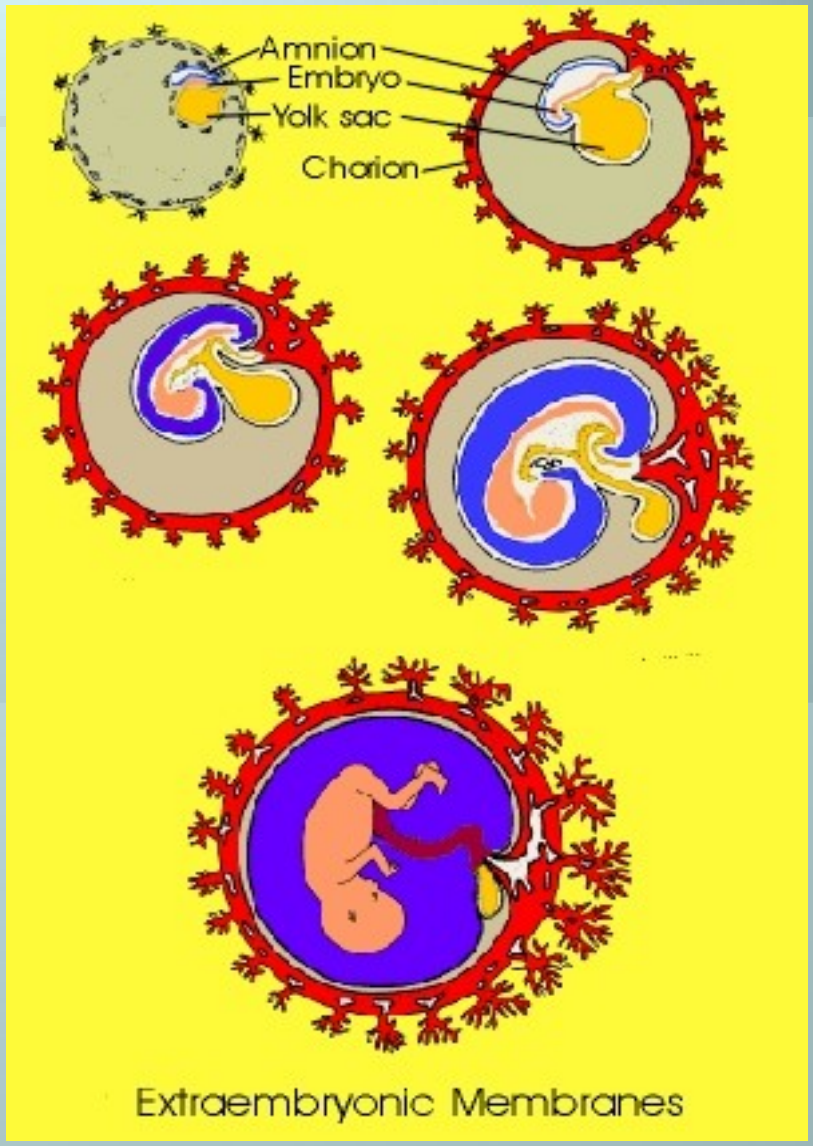


SIGNIFICANCE OF YOLK SAC

- Transfer of nutrients during 2nd and 3rd weeks
- Primordial germ cells appear in endodermal lining of wall of yolk sac in 3rd week
- Incorporate into endoderm of embryo as a primordial gut
- Blood development first occurs here

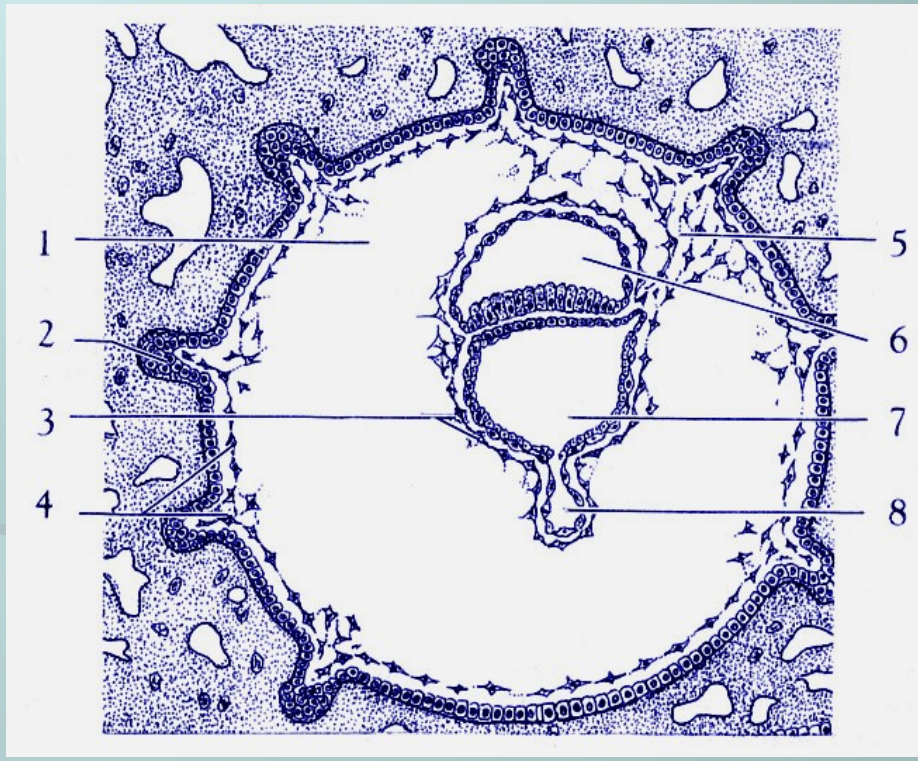
FATE OF YOLK SAC

- At 10 weeks lies in chorionic cavity between chorionic and amniotic sac
- Atrophies as pregnancy advances
- Sometimes it persists throughout pregnancy but of no significance
- In about 2% of adults the proximal intra-abdominal part of yolk stalk persists as an ileal diverticulum or Meckel's diverticulum



ALLANTOIS

- Endodermal finger-like outpouching from caudal part of secondary yolk sac projecting in connecting stalk



- In human embryos, allantois is rudimentary (partly involved in early blood formation, partly associated with development of urinary bladder)

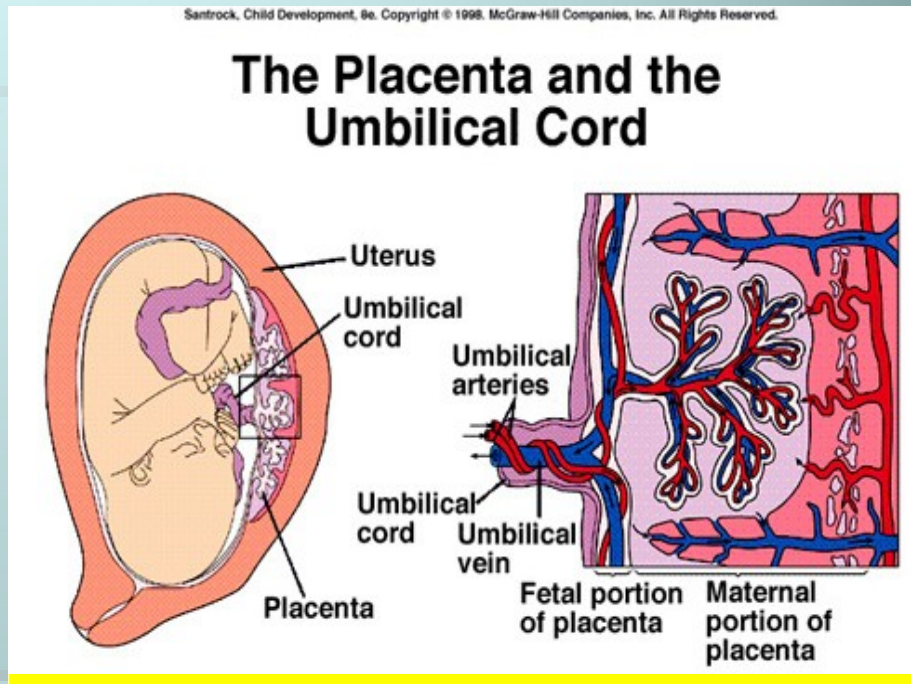
- As bladder enlarges, allantois becomes urachus

FUNCTIONS OF ALLANTOIS

- Blood formation occurs in wall during 3rd to 5th week
- Blood vessels persist as umbilical vein and arteries
- Fluid from amniotic cavity diffuses into umbilical vein and enters fetal circulation for transfer to maternal blood through placental membrane
- Becomes Urachus and after birth is transformed into median umbilical ligament extends from the apex of bladder to the umbilicus

Umbilical Cord

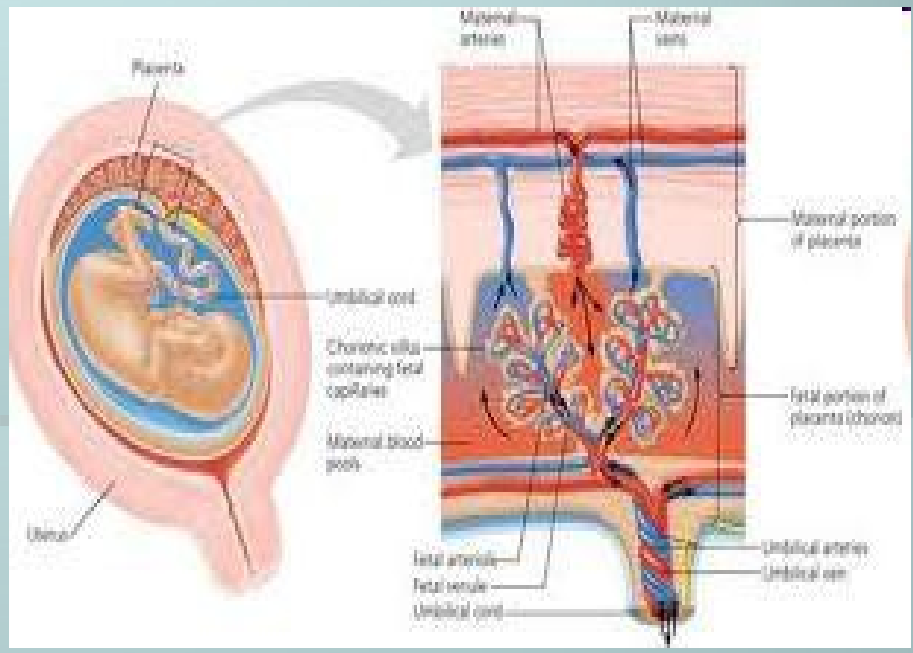
- Attaches fetus to placenta
- Folding – a purse string closure
- Covered with Amnion membrane
- Cord: mucous CT (Wharton's Jelly), 3 vessels, yolk sac, allantois



- At full time, 50-60 cm –it is about equal to length of fetus (as a rule)
- 2cm diameter

DEVELOPMENT OF UMBILICAL CORD

- Developed from yolk sac and allantois
- Replaces yolk sac for supply of nutrients to fetus by end of 5th week
- Not directly connected to mother but connected to placenta
- Blood flow through umbilical cord increases as size of fetus increases

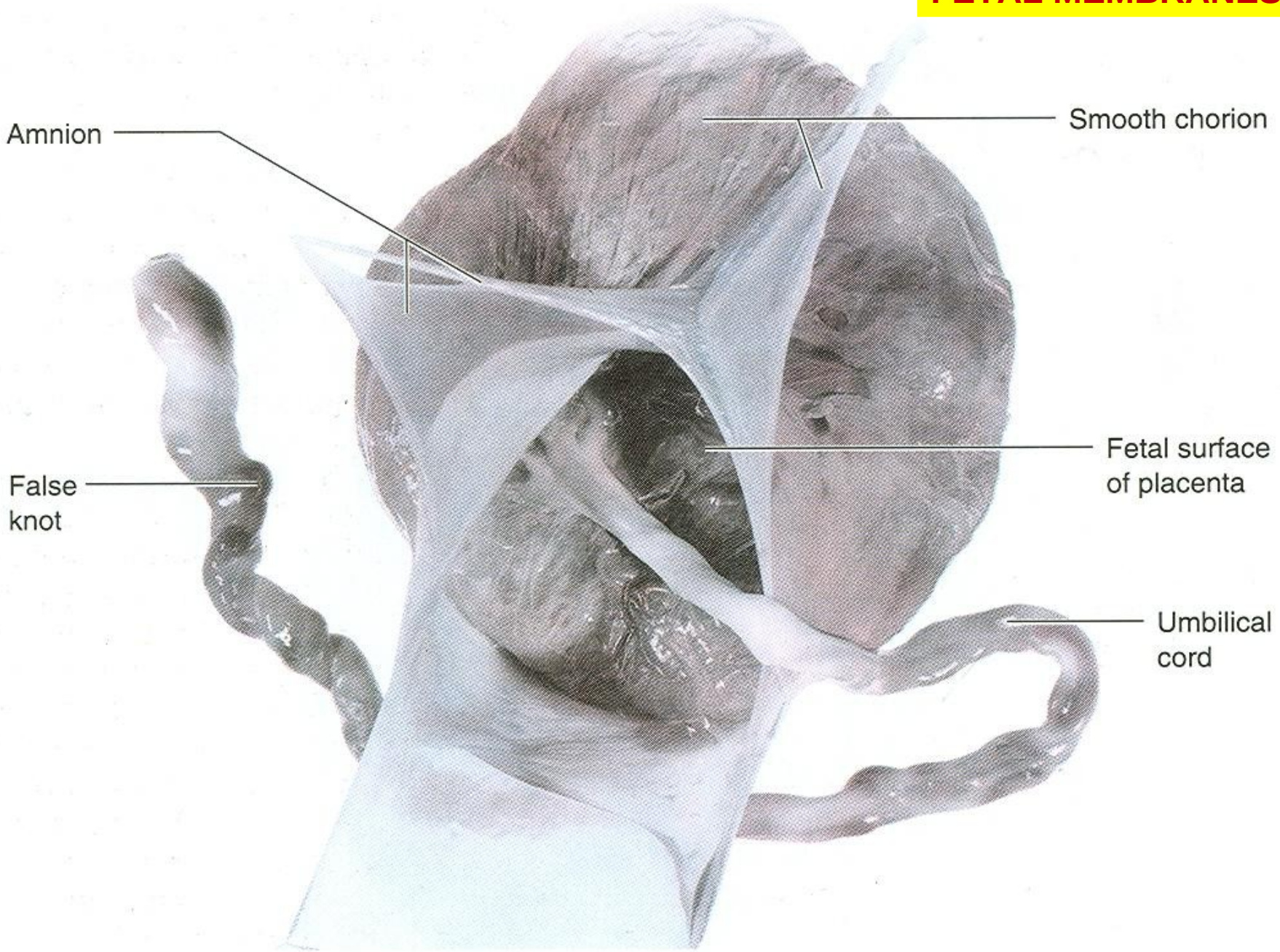


UMBILICAL CORD

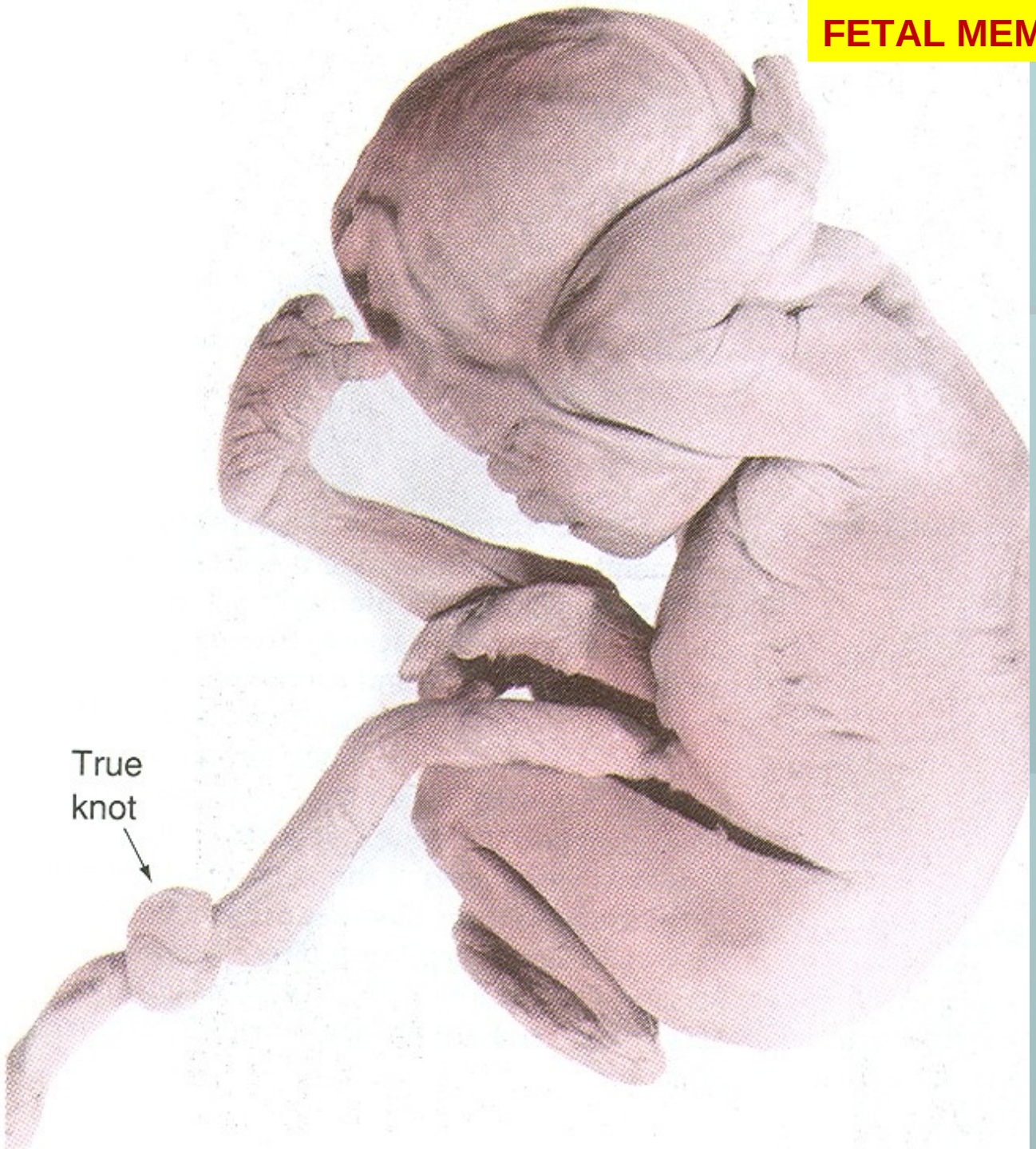
- Umbilical vessels are longer than the cord, so twisting and bending of the vessels are common
- They frequently form loops, producing false knots, that are of no significance
- In about 1% of pregnancies, true knots form in the cord and cause fetal death

APPLIED ANATOMY

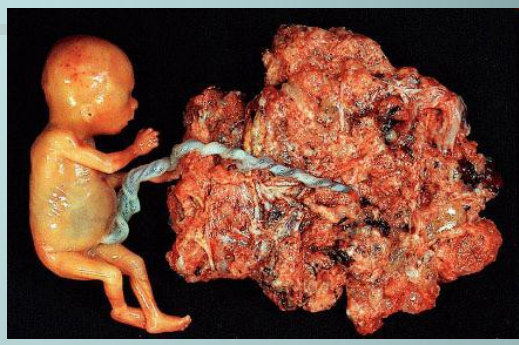
FETAL MEMBRANES



FETAL MEMBRANES



HYDATIDIFORM MOLE

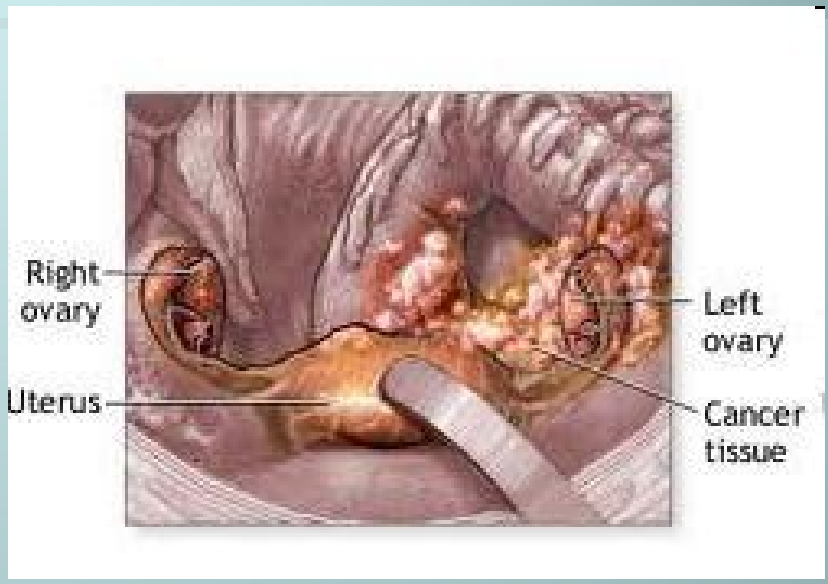


- Overgrowth of placental tissue or an abnormal growth
- Develops from a non viable, fertilized egg

  at the beginning of a pregnancy.
- Referred to as a molar pregnancy.
- Instead of the normal embryonic cell division that results in the development of a fetus, the placental material grows uncontrolled and develops into a shapeless mass of watery, small, blister-like sacs (vesicles).
- The cause of hydatidiform mole is unknown, but is thought to be caused in part by chromosomal abnormalities.

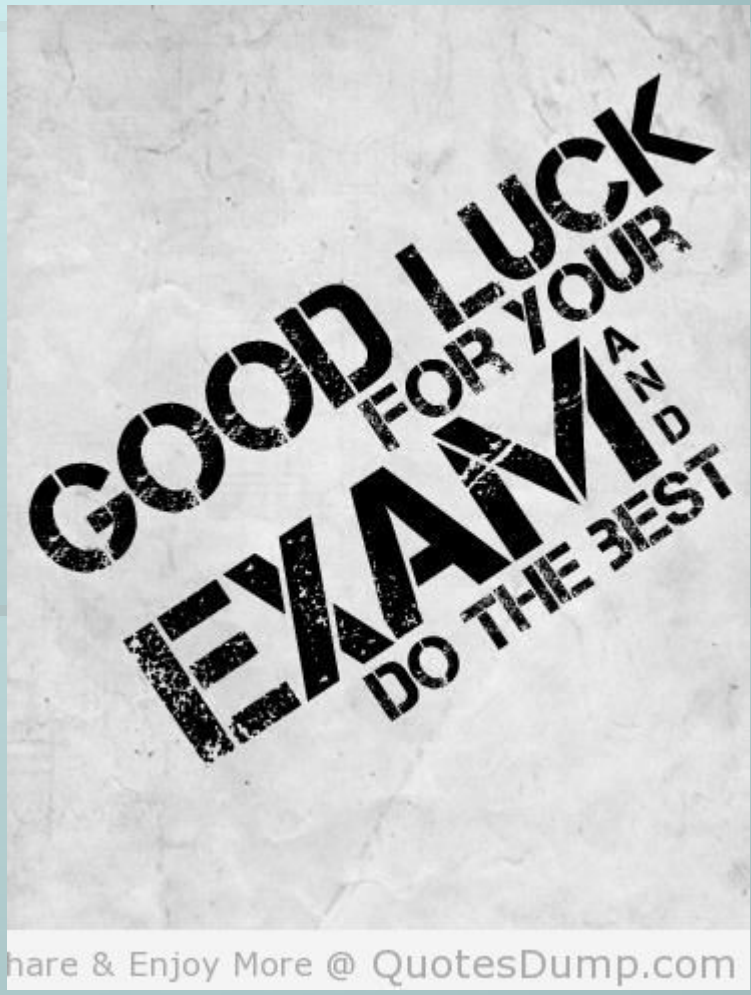
CHORIOCARCINOMA

- Malignant, trophoblastic and aggressive cancer, usually of placenta (malignant end of the spectrum in gestational trophoblastic disease (GTD)
- Characterized by early hematogenous spread to the lungs
- Classified as a germ cell tumor and may arise in the testis or ovary



Characteristic feature is the identification of intimately related syncytiotrophoblasts and cytotrophoblasts without formation of definite placental type villi.


All
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